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SVAR Identification via Changes in Volatility:

A Perturbation Bound for the Problem of Omitted Variables

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Abstract

The present work studies the consequences of omitting relevant DGP variables on structural identification of VAR models via the Changes in Volatility method (CV-SVARs). After reviewing the framework's foundations and the CV identification procedure - which reduces the problem to an eigenvalue decomposition in the whitened space - we analytically prove the inconsistency of the CV estimator in the presence of relevant misspecification. The identification bias is decomposed exactly into two distinct channels: a distortion induced by the Cholesky factor, provoked by contaminations in the first regime, and an eigenvector rotation in the whitened space, driven by regime-2 perturbation. Applying the theorem by Davis and Kahan (1970) in the form of Yu, Wang, and Samworth (2015), we derive a global bound on angular bias as a function of the perturbation norm and the spectral gap of the correctly specified whitened matrix. The spectral gap, coherently with the literature, emerges simultaneously as a source of identifying power and of robustness to incorrect specification. Theoretical predictions are then validated through Monte Carlo experiments and a short empirical application on Italian macro data for the period 1999 - 2019, structured around the volatility regime shift induced by the ECB's 2012 OMT announcement.

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1 Vector Autoregressions and their Structural form

The vector autoregression (VAR) model was introduced by Sims (1980) as a systematic critique of the large-scale structural macroeconomic models that dominated empirical work in the 1970s. Sims argued that the identification restrictions imposed in dynamic simultaneous equation models (DSEMs) - the so-called "Cowles Commission models" (Guala, 2006) - were "incredible": arbitrary exclusion restrictions and normalisation conventions imposed for econometric convenience rather than theoretical necessity. His critique was particularly effective with respect to the treatment of exogeneity. In large macroeconomic models, many variables were assumed exogenous in principle rather than on the basis of sound theoretical reasoning, while policy variables - which on the contrary should have a non-negligible endogenous component - were often treated as if they were externally determined (Sims, 1980). When rational expectations and intertemporal optimisation are taken seriously, especially following the critique of Lucas (1976), the search for truly exogenous instruments becomes even more elusive.

The traditional approach to identification in DSEMs (Lütkepohl and Kilian, 2017, Chap. 6) notoriously sought to pin down structural parameters by imposing exclusion restrictions on the contemporaneous effect matrix Γ and the matrix of lagged effects B , together with instrumental variable methods. To identify a bivariate model of output and money, for instance, the researcher needed at least two exogenous variables - one strongly correlated with demand but not monetary policy, another correlated with monetary policy but not demand (Hamilton, 1994, p. 245). Such instruments were difficult to find and their validity was almost always contestable. Restrictions on lagged endogenous variables, which are predetermined and could in principle be treated as exogenous, were difficult to justify economically since theory provides little guidance on dynamic relationships between variables (Gottschalk, 2001).

By treating all variables symmetrically as endogenous, allowing each one to depend on its own lags and the lags of all others, and imposing no *a priori* restrictions on contemporaneous relationships, Sims's proposal revolutionised past best practices. In his framework, the reduced-form VAR acts as a summary of the data's sampling information. Structural analysis is a subsequent, independent step, achieved by constraining the relationship between residuals and innovations. This view represents a fundamental departure from the traditional modeling practice: whereas DSEMs model the behaviour of variables in levels, SVARs decom-

pose variables into their expected and unexpected components, and model the unexpected part.

This philosophy, which by some authors has been described as agnostic (Uhlig, 2005), sets VARs apart from two other major classes of multivariate time series models in macroeconomics. Dynamic stochastic general equilibrium (DSGE) models derive their empirical implications from first principles of optimising behaviour and market clearing, so their structural parameters carry deep economic interpretations. A development that should invalidate the critique of Lucas and crucially ensure time-invariance of the deep parameters. DSEMs, as noted above, impose instead contemporaneous causal structure through exclusion restrictions on the matrix of simultaneous effects, a strategy also prone to Sims's critique. VARs can be seen as representing a middle ground: they impose minimal structure on the data, making them robust to misspecification of dynamic relationships, while their structural extensions (SVARs) allow economically motivated restrictions to be imposed in a transparent and testable way - although naive structural analysis is in no way exempt from circularity criticism. Where DSEMs can be considered to be primarily designed for policy simulation - i.e., computing dynamic multipliers which describe the impact of an anticipated change in a policy instrument on output - SVARs are designed for the analysis of transmission mechanisms, tracing how the economy responds to an unexpected shock. This distinction has practical implications: impulse response functions, the main output of SVAR analysis, measure the effects of unanticipated discretionary policy actions in the sense of Christiano, Eichenbaum, and Evans (1999, p. 6); hence, they are not directly comparable to the dynamic multipliers obtained from structural models.

The practical success of VARs in macroeconomics ultimately rests on their versatility in performing a broad range of activities (Stock and Watson, 2001): (i) data summarisation; (ii) forecasting and approximation; (iii) structural analysis and causal inference; (iv) policy analysis. Our present work shall mainly concern the third objective, which is why we now need to address the issue of transforming a VAR model into its structural form.

1.1 The Reduced-Form VAR

1.1.1 Model Specification

Let $y_t = (y_{1t}, \dots, y_{Kt})'$ be a K -dimensional stochastic process observed at $t = 1, \dots, T$, where T is the sample dimension. A vector autoregression of order p , $\text{VAR}(p)$, is:

$$y_t = \nu + A_1 y_{t-1} + A_2 y_{t-2} + \dots + A_p y_{t-p} + u_t, \quad (1.1)$$

where $\nu \in \mathbb{R}^K$ is a vector of deterministic intercepts, $A_i \in \mathbb{R}^{K \times K}$ are autoregressive coefficient matrices, and u_t is a vector of reduced-form disturbances - in the form of "one-step-ahead" errors - satisfying:

$$\mathbb{E}[u_t] = 0, \quad \mathbb{E}[u_t u_t^T] = \Sigma_u > 0, \quad \mathbb{E}[u_t u_s^T] = 0 \quad \text{for } t \neq s. \quad (1.2)$$

Thus, they have mean zero and they are not serially correlated. The covariance matrix Σ_u is symmetric positive definite. Its off-diagonal elements reflect contemporaneous correlations among the reduced-form residuals, which cannot be explained by the lagged structure. It is important to stress that these correlations do not, by themselves, have a structural interpretation: u_t is merely a linear combination of the underlying structural innovations, $u_t = B_0^{-1} \varepsilon_t$, and lacks economic content until the structural model is identified (Gottschalk, 2001).

It is possible to use the lag operator L^1 such that $L^i y_t = y_{t-i}$, to compactly write the model as:

$$A(L) y_t = \nu + u_t, \quad A(L) \equiv I_K - A_1 L - A_2 L^2 - \dots - A_p L^p. \quad (1.3)$$

The deterministic component can be extended to include a linear trend or seasonal dummies when appropriate (Lütkepohl and Kilian, 2017). However, the baseline specification (2.1) with intercept only is the most common choice in applied work, with ν being composed of constant terms.

1.1.2 Stability and the Companion Form

The stability properties of the $\text{VAR}(p)$ are most conveniently analysed via the *companion form*, which actually is just a clever algebraical trick to express a p th-order autoregression in the form of an AR(1). Define the Kp -dimensional vector $Y_t = (y_t^T, y_{t-1}^T, \dots, y_{t-p+1}^T)$.

¹ For a thorough discussion on lag operators, see Hamilton (1994, Ch. 2).

Then:

$$Y_t = \mathcal{A}Y_{t-1} + \mathcal{U}_t, \quad \mathcal{A} = \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_K & 0 & \cdots & 0 & 0 \\ 0 & I_K & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_K & 0 \end{bmatrix}. \quad (1.4)$$

The process y_t is *stable* if and only if all eigenvalues of $\mathcal{A}$ lie strictly inside the unit circle:

$$\det(I_{Kp} - \mathcal{A}z) \neq 0 \quad \text{for all } |z| \leq 1. \quad (1.5)$$

Stability implies covariance stationarity and is equivalent to requiring that the effect of any shock dissipates over time. Intuitively, when a stochastic process is not stable, past shocks accumulate rather than die out with the passing of time, leading to asymptotic explosive behaviour. Under stability, though, the process is guaranteed to admit what is referred to as the Wold moving average representation (Pfaff, 2008, pp. 24-25):

$$y_t = \nu^* + \sum_{i=0}^{\infty} \Phi_i u_{t-i}, \quad \Phi_0 = I_K, \quad A_j = 0 \leftrightarrow j > p \quad (1.6)$$

where $\nu^* = A(1)^{-1}\nu$ and the coefficient matrices Φ_i decay geometrically. They are recovered via the recursion:

$$\Phi_s = \sum_{j=1}^s \Phi_{s-j} A_j \quad s \geq 1. \quad (1.7)$$

The MA representation is central to impulse response analysis: the element $[\Phi_s]_{lj}$ gives the response of variable l at horizon s to a unit innovation in the j -th reduced-form residual at time t , holding all other innovations at all dates constant (Hamilton, 1994, p. 318). It is this representation that the SVAR approach then exploits to attach structural economic content to the innovations.

1.1.3 Lag Order Selection and Granger Causality

The choice of lag order p involves a tradeoff between variance and bias (De Graeve and Westermarck, 2025). An underspecified model introduces bias and residual autocorrelation, leading to inconsistent impulse response estimates. Each lag dynamic not explicitly modelled pollutes the residuals with its recursive behaviour. An overspecified model, on the other hand, usually inflates the variance of estimation and reduces the precision of inference. In the SVAR framework, it is assumed that no restrictions are imposed on the lag dynamics: all lagged

variables are included in every equation, so that the coefficient matrices $A_1, \dots, A_p$ are left unrestricted (Gottschalk, 2001). This is in deliberate contrast to DSEMs, where restrictions on lagged effects are common but hard to justify theoretically (see previous discussion). An obvious problem shows as soon as we try to decide the VAR order. How many lags should we include for each variable to be sure of capturing its principal regressive dynamics, without incurring in excessive overfitting? The traditional wisdom has been to keep lag horizon low, in order to keep inference precise and the model manageable; however, there have recently been some contrarian opinions (De Graeve and Westermarck, 2025).

For most applications, lag order selection is typically based on information criteria of the form:

$$\text{IC}(p) = \log \det \hat{\Sigma}_u(p) + c_T \cdot \frac{K^2 p}{T}, \quad (1.8)$$

where $\hat{\Sigma}_u(p)$ is the estimated covariance matrix under order p and c_T is a penalty term. The most commonly used criteria are:

$$\text{AIC}(p) = \log \det \hat{\Sigma}_u(p) + \frac{2}{T} K^2 p, \quad (1.9)$$

$$\text{HQ}(p) = \log \det \hat{\Sigma}_u(p) + \frac{2 \log \log T}{T} K^2 p, \quad (1.10)$$

$$\text{BIC}(p) = \log \det \hat{\Sigma}_u(p) + \frac{\log T}{T} K^2 p. \quad (1.11)$$

The AIC tends to overestimate the true lag order asymptotically, while the BIC and HQ criteria are consistent under standard regularity conditions (Lütkepohl, 2005). In practice, the selected lag order should be verified by testing for residual autocorrelation via the multivariate Breusch-Godfrey LM test. Another common test for residuals uncorrelatedness is the Portmanteau test. For quarterly data, lag orders of two to four are typical; for monthly data, from six to twelve, to reflect the appropriate frequency of the data.

An important application of reduced-form VARs that does not require structural identification is the analysis of Granger causality (Granger, 1969). Variable y_{1t} *Granger causes* y_{2t} if past values of y_{1t} contain predictive information about y_{2t} beyond the own past of y_{2t} . In the VAR framework, y_{1t} does not Granger cause y_{2t} if and only if the relevant submatrices $[A_i]_{2,1} = 0$ for all $i = 1, \dots, p$, a restriction testable by a standard Wald test - similar to a standard F test - on the OLS estimates (Pfaff, 2008). Granger causality is distinct from structural causality: it may show that money leads output in time, but this does not establish that money causes output, since monetary authorities may systematically anticipate

future output movements and adjust policy accordingly (Hamilton, 1994, p. 305). Structural causality requires the identification of unexpected shocks, which is precisely the task of the SVAR framework. Incidentally, Granger causality tests might also informally serve as a diagnostic for potential omitted variable bias: if a possible omitted variable strongly Granger causes the other variables in the model, its exclusion is likely to contaminate the residuals and distort structural inference. The problem of omitted variable bias will be investigated in more depth in the next sections of our work.

1.2 From Reduced-Form to Structure

The reduced-form VAR describes the statistical properties of y_t efficiently, but its disturbances u_t are not directly interpretable as economic shocks. This is precisely what makes VAR an excellent tool for forecasting and summarising the data, but a poor instrument to perform causal inference (i.e., trying to recover the underlying economic structure that generated the data themselves). Structural innovations must be understood as exogenous forces, not directly observed by the econometrician, which strike the system and cause oscillations (Christiano, Eichenbaum, and Evans, 1999). Because these shocks are exogenous, i.e., they do not have common causes, it is reasonable to address them as uncorrelated. This orthogonality of structural innovations is the key assumption that distinguishes the SVAR approach from the traditional simultaneous equations approach, where the variance-covariance matrix of structural disturbances was left unrestricted, as these latter were thought as merely representing errors in the equations (Gottschalk, 2001).

The structural form is:

$$B_0 y_t = \nu_0 + B_1 y_{t-1} + \cdots + B_p y_{t-p} + \varepsilon_t, \quad (1.12)$$

where B_0 is an invertible matrix of contemporaneous effects and ε_t are the structural shocks satisfying:

$$\mathbb{E}[\varepsilon_t] = 0, \quad \mathbb{E}[\varepsilon_t \varepsilon_t^T] = \Sigma_\varepsilon, \quad \mathbb{E}[\varepsilon_t \varepsilon_s^T] = 0 \quad \text{for } t \neq s. \quad (1.13)$$

Pre-multiplying by B_0^{-1} gives the reduced form with $u_t = B_0^{-1} \varepsilon_t$. Different normalisation conventions for Σ_ε give rise to distinct SVAR parameterisations, as discussed below.

The SVAR's ‘‘shock view’’ can be derived directly from the structural form, because the latter now includes the contemporaneous effects coefficients. Subtracting from both sides of (1.12) the conditional expectation $\mathbb{E}_{t-1}[B_0 y_t]$, the deterministic lagged components drop out and

one is left with:

$$B_0 u_t = \varepsilon_t, \quad (1.14)$$

where $u_t = y_t - \mathbb{E}_{t-1}[y_t]$ is the vector of reduced-form forecast errors. Equation (1.14) is the fundamental relationship of SVAR analysis: it expresses the structural innovations as a linear transformation of the observable forecast errors. This also clarifies the normalization: SVAR models seek to estimate the impulse response functions showing the response to a standard-deviation shock, which motivates setting $\mathbb{E}[\varepsilon_t \varepsilon_t^T] = I_K$, so that unit innovations in ε_t correspond to one-standard-deviation shocks.

1.2.1 The identification problem and the CV method

By knowing B_0 , one can recover the structural shocks from the reduced-form residuals. The fundamental issue with SVAR identification is that, to obtain a unique impact matrix, we would need K^2 free parameters; unfortunately, though, the reduced-form covariance matrix Σ_u only has $K(K+1)/2$ free parameters as it is a symmetric matrix. To see why this is actually a problem, consider a possible impact matrix for which $\Sigma_u = (B_0 B_0^T)^{-1}$. Now, take a different matrix $B' = B_0^{-1}Q$, where Q is any orthogonal matrix (i.e., $QQ^T = I$ or equivalently $Q^{-1} = Q^T$). It is easy to show that $BB^T = B_0^{-1}QQ^TB_0^{-T} = (B_0 B_0^T)^{-1} = \Sigma_u$. In words, any rotation of B_0^{-1} would actually serve our purpose. To obtain point identification, then, we usually need to impose $K(K-1)/2$ further restrictions on the impact matrix (Pfaff, 2008). These restrictions are not invariant under rotation: a generic $Q \neq I$ will violate them, and if sufficiently many independent restrictions are imposed, the only admissible rotation is the identity.

The traditional methods for imposing restrictions are grounded in economic theory. For instance, the recursiveness assumption imposes *a priori* a causal ordering on the system's variables (Christiano, Eichenbaum, and Evans, 1999), as follows: the first variable responds contemporaneously only to its own shock, the second to the first and its own, and so on². The ordering can be imposed either on each variable or on blocks of "sister" variables: the only constraint is that the resulting matrix be lower triangular, as in a Cholesky decomposition. Indeed, this imposes $K(K-1)/2$ zero restrictions – precisely the number needed for identification.

² To provide a concrete example: in a monetary policy context, this would mean that the central bank cannot respond instantaneously to developments in the real economy due to lags in data collection, so that the monetary policy shock does not affect output and prices within the period.

Other common identification schemes include *short-run* restrictions, which are zero restrictions placed on specific entries of the impact matrix in accordance with the economic literature; *long-run* restrictions, pioneered by Blanchard and Quah (1988), which constrain the long-horizon dynamics of the resulting impulse response functions; and *sign restrictions* (Uhlig, 2005), which impose only the sign pattern of the impulse responses over a specified horizon, leaving coefficient magnitudes unrestricted³. More recently, the use of external instruments has become mainstream in the SVAR literature. First introduced by Stock in 2008 and subsequently formalised by Stock and Watson (2012) and Mertens and Ravn (2013), this approach identifies structural shocks by exploiting variables that are correlated with the shock of interest but orthogonal to all remaining shocks – the macroeconomic counterpart of instrumental variables in microeconometrics.

All the aforementioned approaches rely, to varying degrees, on economic restrictions established *ex ante*. An alternative strategy is to exploit some statistical properties of the data themselves — in our case of interest, heteroskedasticity of the structural shocks — to achieve full or partial identification without imposing arbitrary restrictions on the impact matrix. This is the idea underlying the statistical, or data-driven, methods of identification, which include the *Change in Volatility* (CV) technique. This latter, rather than importing identifying information from outside the model, extracts it from a second-moment statistical property of the data; namely, the heteroskedasticity of the structural shocks. The first to formalize the importance of volatility discrepancies among variables was Fisher (1965), in the context of systems of structural equations. The approach was then recovered and applied to the VAR framework by Sentana and Fiorentini (2001). However, the most immediate intuition can actually be found in Rigobon (2003): if the variances of the structural shocks change across sub-periods, while the impact matrix B remains **constant** throughout the sample, then each distinct volatility regime delivers an additional covariance matrix, and each such matrix adds $K(K + 1)/2$ moment equations to the identification system at the cost of introducing only K new variance parameters (one per shock), effectively providing the $K(K + 1)/2 - K = K(K - 1)/2$ additional parameters needed for identification. The K new parameters are simply the *volatility ratios* expressing the relative change in volatility between regimes for each variable. The identification mechanism relies on the fact that time series periods with different volatilities of the system’s variables allow to compute distinct

³ Sign restrictions are sometimes preferred because they are perceived as more “agnostic”, in the sense of carrying fewer implicit assumptions. However, they identify only a set of admissible rotations of the impact matrix rather than a unique one, and are therefore generally insufficient for point identification.

covariance matrices; having two or more of these is in fact sufficient to obtain enough free parameters to estimate the K^2 entries of B_0 .

A fundamental prerequisite of this identification method is that each volatility ratio must be distinct from the others in order to obtain full identification (if this fails, partial identification could still be possible for those shocks satisfying the condition)⁴. The distinctness condition has a transparent economic interpretation: it fails if and only if two structural shocks change their relative variances by the same factor across regimes. Although in applications this is often a rather innocuous hypothesis in the presence of abrupt shifts in the economic system - such as financial crises, wars, and monetary regime disruptions just to name a few - not all macroeconomic datasets contain a sufficient degree of heteroskedasticity to permit the employment of this identification method. The researcher should always exercise caution in testing the hypotheses of effective structural breaks (Lütkepohl et al., 2021), and only adopt CV when its preconditions are clearly met by the data.

Rigobon (2003) focuses on a single pair of distinct volatility regimes; however, the case of $M > 2$ regimes generalises naturally. Each additional regime contributes a further covariance matrix $\Sigma^{(m)} = B\Lambda^{(m)}B^T$ with K new unknown diagonal elements $\lambda_j^{(m)}$, but the K^2 elements of B remain the same across all regimes. The identification condition for the M -regime case is that for every pair (j, l) with $j \neq l$, there exists at least one regime m such that $\lambda_j^{(m)} \neq \lambda_l^{(m)}$ (Lütkepohl and Kilian, 2017, p. 497). The usual requirement might become easier to satisfy as M increases. A further implication of this methodology is worth to stress: once B is recovered from the heteroskedastic structure alone, *any* conventional identifying restriction - a specific zero pattern, or perhaps an hypothesized recursive ordering - becomes an over-identifying restriction that can be tested by a standard likelihood-ratio statistic against the null hypothesis that the restriction is implied by the data-identified B (Lanne and Lütkepohl, 2008, p. 1142). The heteroskedastic model employed in this fashion does not replace economic identification; rather, it makes it empirically testable. Competing structural interpretations that previously had to be trusted on faith can be assessed against the data, and a scheme that is rejected by the volatility-based over-identification test can be regarded as incompatible with the observed second-moment dynamics.

Before proceeding, a notational remark is in order. This section introduced the structural form in the A-model convention, following Lütkepohl and Kilian (2017), writing $B_0 y_t =$

⁴ We do not discuss this matter here for reasons of space. A very clear explanation can be found in Lütkepohl and Kilian (2017, p. 494). When volatility ratios are not distinct, this situation is referred to as *weak identification*. The following section of our work tries to give a geometric explanation of this phenomenon.

$\dots + \varepsilon_t$ so that $u_t = B_0^{-1}\varepsilon_t$ and $\Sigma_u = (B_0 B_0^T)^{-1}$. From the next section onwards, following standard practise (Lütkepohl, 2013), the analysis is conducted in the B-model, where $B \equiv B_0^{-1}$ is the structural impact matrix:

$$u_t = B\varepsilon_t, \quad \Sigma_u = B\Sigma_\varepsilon B^T. \quad (1.15)$$

The two parameterisations are algebraically equivalent and carry identical empirical content. The B-model is preferred here because the simultaneous diagonalization and the eigenvalue decomposition of Section 2 are stated most cleanly in terms of the impact matrix directly. One final concern deserves mention, in order to properly introduce the problematic discussed in the section that follows. The identification result above is an asymptotic statement: given consistent estimates of $\Sigma^{(1)}$ and $\Sigma^{(2)}$, the decomposition recovers B exactly in the population. In finite samples, the recovered impact matrix might substantially differ from the one constituting the underlying data generating process. However, the present work is concerned with a prior source of fragility in CV identification: the consequences of working with a misspecified system, from which one - or possibly more - relevant variables have been omitted. As we show, omitted variable bias interacts with the eigenvalue structure of the whitened covariance matrix in a way that is directly related to the spectral gap between the volatility ratios.

2 Inconsistency of CV identification due to the omission of relevant variables

The fact that identification deteriorates in the presence of omitted variables is rather pacific in the SVAR literature (Lütkepohl and Kilian, 2017). However, it is usually not discussed with the aid of theoretic arguments in the context of the CV identification. Thus, the aim of this section is to present a formal account of the phenomenon, by providing an inconsistency proof, a bias decomposition and a useful tool, derived from the theory of matrix perturbations, to interpret the conditions under which shocks are more robust to omitted variable bias (OVB). The theorem in question is the one by Davis and Kahan (1970). The originality of the present work precisely lies in applying a well-known theorem in Mathematics and Statistics to the specific context of SVAR models, in the hope of achieving some intuitive insights on the dynamics of the perturbation. It is also instructive to connect our results on weak identification to the relevant literature: the role of variance dispersion in structural identification has been studied from three complementary perspectives. Fisher (1965) provided the first rigorous formalisation of the classical intuition that disparity in the variances of structural disturbances can aid identification: in a simultaneous system, the equation with the smallest disturbance variance is nearly identified precisely because the large shifts of the other equations trace it out. Variance dispersion, sufficiently heterogeneous, can thus become a main source of identification. Lewis (2022) examines the same dispersion from the inferential side: when variance shifts across regimes are small or proportional, the identifying variation vanishes and standard inference breaks down – dispersion that is insufficient becomes then a *problem* for identification. The present work studies a third dimension of the same phenomenon: even when identification is nominally achieved, the omission of a relevant variable contaminates the recovered impact matrix in a way that is governed by the very same spectral structure. Variance dispersion thus also determines the *frailty* of identification to misspecification, with the spectral gap δ_j (defined in the manner that will follow) acting simultaneously as the source of identifying power and as the index of resilience to omitted variable bias.

2.1 Identification via Changes in Volatility (two regimes)

In this section, we will present the identification procedure used in the Change of Volatility SVAR framework. The approach relies on transforming the problem into the whitened space

via a Cholesky decomposition. From there, the structural shocks can be recovered through a generalized eigenvalue decomposition. The procedure presented, which is the product of the authors' personal reframing coherently with our objectives, is - crucially - equivalent to the one described in Lewis (2025, p. 670), as well as to the one implemented via *id.cv()* from the **svars** package in R (Lange et al., 2021)⁵. Although it is perfectly possible to envision scenarios with more than one structural breaks - and thus more than two volatility regimes -, we focus on the simpler case of Rigobon (2003) to keep the topics at a more tractable level.

Structural Setup

Consider a $\text{VAR}(p)$ representation of the data-generating process

$$y_t = A_1 y_{t-1} + \cdots + A_p y_{t-p} + B w_t, \quad (2.1)$$

where

- $y_t \in \mathbb{R}^n$ is the vector of observable variables,
- $B \in \mathbb{R}^{n \times n}$ is the structural impact matrix,
- w_t is a vector of structural shocks.

The structural shocks are assumed to be mutually uncorrelated, but exhibit regime-dependent variances:

$$\text{Var}(w_t \mid k) = \Lambda^{(k)} = \text{diag}\left(\lambda_1^{(k)}, \dots, \lambda_n^{(k)}\right), \quad \lambda_j^{(k)} > 0. \quad (2.2)$$

The diagonal structure of $\Lambda^{(k)}$ reflects the assumption that structural shocks are orthogonal. Identification relies on the fact that their variances change distinctly across regimes. As we have seen in Section 1, this is an uncontroversial assumption in the SVAR framework.

The reduced-form residuals are given by $u_t = B w_t$. Therefore, their covariance matrix in regime k is

$$\Sigma^{(k)} = \text{Var}(u_t \mid k) = B \Lambda^{(k)} B^T \quad (2.3)$$

Following the standard normalisation in heteroskedastic SVARs, we set

$$\Lambda^{(1)} = I.$$

⁵ Verified numerically on a synthetic DGP.

Under this normalisation,

$$\Sigma^{(1)} = BB^T.$$

This artifice fixes the scale of the structural shocks in the reference regime and implies that the covariance matrix in regime 1 is purely determined by the structural impact matrix. Furthermore, it allows us to simplify the analysis because the volatility ratios $\lambda_j^{(2)}/\lambda_j^{(1)} \neq \lambda_i^{(2)}/\lambda_i^{(1)}$ are now expressed solely in terms of $\lambda_j^{(2)} \neq \lambda_i^{(2)}$.

Cholesky Decomposition and Whitening Transformation

Solving the identification problem requires finding a matrix B which simultaneously diagonalizes both our variance-covariance matrices. In order to do this, we exploit the fact that $\Sigma^{(1)}$ is symmetric positive definite (SPD) and admits a Cholesky factorisation

$$\Sigma^{(1)} = CC^T, \quad (2.4)$$

where C is a lower triangular matrix with positive diagonal entries. While C would actually be sufficient to recover a potential impact matrix, its lower triangular structure irremediably imposes a recursive ordering on the system (Lütkepohl and Kilian, 2017). Since such an assumption is often economically unjustifiable as discussed in Section 1.2, we now seek an orthogonal matrix Q to rotate C such that it simultaneously diagonalizes the second variance-covariance matrix, $\Sigma^{(2)}$.

The identification problem can thus be expressed as a *generalised eigenvalue problem* (GEP). We seek vectors v and scalars λ satisfying

$$\Sigma^{(2)}v = \lambda \Sigma^{(1)}v = \lambda CC^T v. \quad (2.5)$$

Multiplying both sides on the left by C^{-1} yields

$$C^{-1}\Sigma^{(2)}v = \lambda C^T v. \quad (2.6)$$

Define the change of variable $\omega = C^T v$, so that $v = C^{-T}\omega$. Substituting,

$$C^{-1}\Sigma^{(2)}C^{-T}\omega = \lambda \omega. \quad (2.7)$$

This change of variable maps the generalised eigenvalue problem into a standard eigenvalue problem. The whitened matrix

$$W = C^{-1}\Sigma^{(2)}C^{-T} \quad (2.8)$$

is SPD (since also $\Sigma^{(2)}$ is SPD), so all its eigenvalues are real and positive and its eigenvectors are orthonormal.

After whitening, the first covariance matrix becomes the identity, while the second covariance matrix contains the identifying information.

Recovery of the Structural Impact Matrix

Solving for the eigenvectors of W is straightforward, and yields the following eigen-decomposition:

$$Q^T W Q = \Lambda^{(2)} \quad (2.9)$$

where Q is the orthogonal matrix of eigenvectors (normalized such that $Q^T Q = I$). Substituting the definition of W in the last expression

$$Q^T C^{-1} \Sigma^{(2)} C^{-T} Q = \Lambda^{(2)} \quad (2.10)$$

Now, define $S = C^{-T} Q$. This matrix is exactly the one we need in order to diagonalize both covariance matrices simultaneously. Indeed,

$$S^T \Sigma^{(1)} S = Q^T C^{-1} C C^T C^{-T} Q = I \quad (2.11)$$

and

$$S^T \Sigma^{(2)} S = Q^T C^{-1} \Sigma^{(2)} C^{-T} Q = Q^T W Q = \Lambda^{(2)} \quad (2.12)$$

.

To recover the structural impact matrix B , we need to take the inverse-transpose of S , since this directly yields

$$\Sigma^{(1)} = B B^T, \Sigma^{(2)} = B \Lambda^{(2)} B^T$$

as requested by the definition of u_t .

Summary of the Identification Procedure

To sum up, the identification algorithm can be summarised as follows:

1. Estimate the covariance matrices $\Sigma^{(1)}$ and $\Sigma^{(2)}$.

2. Compute the Cholesky factorisation $\Sigma^{(1)} = CC^T$.
3. Construct the whitened matrix $W = C^{-1}\Sigma^{(2)}C^{-T}$.
4. Compute the eigenvalue decomposition $W = Q\Lambda^{(2)}Q^T$.
5. Recover the structural impact matrix $B = CQ$.

This procedure shows that CV identification reduces to an eigenvalue decomposition in the whitened space, followed by a back-transformation via the Cholesky factor. However, note that B is only identified, as usual, up to sign change and permutation of its columns.

2.2 Setup

For what follows, let the true data-generating process (DGP) be a VAR(2) with $n = 4$ variables:

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + B w_t, \quad (2.13)$$

where $y_t \in \mathbb{R}^4$, $A_1, A_2, B \in \mathbb{R}^{4 \times 4}$, and $w_t \in \mathbb{R}^4$ is a vector of mutually uncorrelated structural shocks. The model exhibits heteroskedasticity across two regimes as in (2.2). Without loss of generality, we normalise the variance of the structural shocks in the reference regime such that $\Lambda^{(1)} = I$. Consequently, the uncontaminated covariance in regime 1 is purely structural: $\Sigma^{(1)} = BB^T$. We ought to stress that none of our results depends on the arbitrarily imposed lag order selection, nor on the number of variables included in the system.

Now, suppose the researcher observes only $\tilde{y}_t \in \mathbb{R}^3$, the first three components of y_t . Partition the DGP conformably:

$$y_t = \begin{pmatrix} \tilde{y}_t \\ y_{4t} \end{pmatrix}, \quad A_i = \begin{pmatrix} \tilde{A}_i & a_i \\ * & * \end{pmatrix}, \quad B = \begin{pmatrix} \tilde{B} & c \\ * & * \end{pmatrix}, \quad (2.14)$$

where $\tilde{A}_i \in \mathbb{R}^{3 \times 3}$, $a_i \in \mathbb{R}^{3 \times 1}$ for $i = 1, 2$, $\tilde{B} \in \mathbb{R}^{3 \times 3}$, and $c \in \mathbb{R}^3$ is the impulse vector of the omitted shock w_{4t} .

The true process for the observed variables is

$$\tilde{y}_t = \tilde{A}_1 \tilde{y}_{t-1} + \tilde{A}_2 \tilde{y}_{t-2} + a_1 y_{4,t-1} + a_2 y_{4,t-2} + \tilde{B} \tilde{w}_t + c w_{4t}, \quad (2.15)$$

whereas the researcher estimates $\tilde{y}_t = \hat{A}_1 \tilde{y}_{t-1} + \hat{A}_2 \tilde{y}_{t-2} + \hat{u}_t$. Provided the VAR is stationary, then, the reduced-form residuals converge in probability to

$$\hat{u}_t \xrightarrow{p} \underbrace{\tilde{B} \tilde{w}_t}_{\tilde{u}_t} + \underbrace{c w_{4t} + a_1 y_{4,t-1} + a_2 y_{4,t-2} + (\tilde{A}_1 - \hat{A}_1) \tilde{y}_{t-1} + (\tilde{A}_2 - \hat{A}_2) \tilde{y}_{t-2}}_{\eta_t}, \quad (2.16)$$

where $\tilde{u}_t$ are the clean residuals and η_t is the total contamination, combining the contemporaneous impact of w_{4t} and its lagged dynamics. The coefficient matrices $\hat{A}_i$ may well not converge to their true values $\tilde{A}_i$, adding to the contamination term. Notice, then, how $\hat{u}_t$ violates the hypothesis of white noise residuals (1.2), due to the serial correlation of the contamination terms. The covariance of $\hat{u}_t$ in regime k converges to

$$\hat{\Sigma}^{(k)} = \tilde{\Sigma}^{(k)} + P_k, \quad (2.17)$$

where $\tilde{\Sigma}^{(k)} = \tilde{B}\tilde{\Lambda}^{(k)}\tilde{B}^T$ is the uncontaminated 3×3 covariance and

$$P_k = \mathbb{E}[\eta_t \eta_t^T \mid k] \quad (2.18)$$

captures both the contemporaneous variance of the omitted shock and the autocovariance structure generated by the lagged dynamics of y_{4t} . We stress that, generically, $P_1 \neq P_2$ because we allow for changes in the variance of the omitted variable. Notice that $\mathbb{E}[\tilde{u}_t \eta_t^T]$ and $\mathbb{E}[\eta_t \tilde{u}_t^T]$ both equal zero because of the definition of the structural shock w ⁶. P_k is SPD under Assumption A.1 also found in the Appendix.

Effect of Misspecification

Following the identification procedure of Section 2.1, the correctly specified 3×3 system would require computing the Cholesky factorisation $\tilde{\Sigma}^{(1)} = \tilde{C}\tilde{C}^T$, constructing the whitened matrix $\tilde{W} = \tilde{C}^{-1}\tilde{\Sigma}^{(2)}\tilde{C}^{-T}$, and solving for its eigenvectors $\tilde{Q}$ via $\tilde{Q}^T\tilde{W}\tilde{Q} = \tilde{\Lambda}^{(2)}$. The structural impact matrix would then be recovered as $\tilde{B} = \tilde{C}\tilde{Q}$, in agreement with Step 5 of the identification algorithm.

Under correct specification, the matrix $\tilde{Q} = \tilde{C}^{-1}\tilde{B}$ is orthogonal. Indeed,

$$\tilde{Q}\tilde{Q}^T = \tilde{C}^{-1}\tilde{B}\tilde{B}^T\tilde{C}^{-T} = \tilde{C}^{-1}\tilde{\Sigma}^{(1)}\tilde{C}^{-T} = \tilde{C}^{-1}\tilde{C}\tilde{C}^T\tilde{C}^{-T} = I, \quad (2.19)$$

so the columns of $\tilde{Q}$ form an orthonormal basis in the whitened space, and the identification procedure of Section 2.1 goes through exactly.

In the presence of misspecification, however, the researcher does not observe $\tilde{\Sigma}^{(k)}$ but only the contaminated matrices $\hat{\Sigma}^{(k)} = \tilde{\Sigma}^{(k)} + P_k$. The researcher therefore computes the contaminated Cholesky factorisation

$$\hat{\Sigma}^{(1)} = \hat{C}\hat{C}^T, \quad \hat{C} \neq \tilde{C}, \quad (2.20)$$

⁶ See Lemma A.1 in the Appendix

and constructs the corresponding whitened matrix

$$\hat{W} = \hat{C}^{-1} \hat{\Sigma}^{(2)} \hat{C}^{-T} = \hat{C}^{-1} (\tilde{B} \tilde{\Lambda}^{(2)} \tilde{B}^T + P_2) \hat{C}^{-T}. \quad (2.21)$$

Remark. Although both $\hat{\Sigma}^{(1)}$ and $\hat{\Sigma}^{(2)}$ are contaminated, the whitened matrix $\hat{W}$ remains symmetric positive definite. Indeed,

$$\hat{W}^T = (\hat{C}^{-1} \hat{\Sigma}^{(2)} \hat{C}^{-T})^T = \hat{C}^{-1} (\hat{\Sigma}^{(2)})^T \hat{C}^{-T} = \hat{C}^{-1} \hat{\Sigma}^{(2)} \hat{C}^{-T} = \hat{W}, \quad (2.22)$$

where symmetry of $\hat{\Sigma}^{(2)} = \tilde{\Sigma}^{(2)} + P_2$ is ensured since it is a second-moment matrix. The property of being SPD follows from invertibility of $\hat{C}$ and the SPD property of $\hat{\Sigma}^{(2)}$. The spectral theorem therefore still applies to $\hat{W}$, and its eigenvectors $\hat{Q}$ are orthonormal by construction. The failure of identification is thus not a failure of the algebraic structure of $\hat{W}$, but rather the consequence of two independent distortion channels introduced by P_1 and P_2 , each sufficient on its own to break identification.

Channel 1 (P_1 corrupts the whitening step). The Cholesky factor $\hat{C}$ whitens $\hat{\Sigma}^{(1)}$, not $\tilde{\Sigma}^{(1)}$. As a consequence, the space whitened by $\hat{C}^{-1}$ is not the one in which the true structural impact matrix $\tilde{B}$ lives. To see this formally, note that under correct specification, $\tilde{Q} = \tilde{C}^{-1} \tilde{B}$ is orthogonal by (2.19), meaning that $\tilde{C}^{-1}$ maps the columns of $\tilde{B}$ into an orthonormal basis. By contrast, consider the image of $\tilde{B}$ under the contaminated whitening $\hat{C}^{-1}$:

$$\hat{Q}_s \equiv \hat{C}^{-1} \tilde{B}. \quad (2.23)$$

This matrix satisfies

$$\hat{Q}_s \hat{Q}_s^T = \hat{C}^{-1} \tilde{B} \tilde{B}^T \hat{C}^{-T} = \hat{C}^{-1} \tilde{\Sigma}^{(1)} \hat{C}^{-T} = \hat{C}^{-1} (\hat{\Sigma}^{(1)} - P_1) \hat{C}^{-T} = I - \hat{C}^{-1} P_1 \hat{C}^{-T} \neq I \quad \text{unless } P_1 = 0. \quad (2.24)$$

Hence $\tilde{B}$ does not lie in the whitened space induced by $\hat{C}$: its columns, once mapped by $\hat{C}^{-1}$, no longer form an orthonormal basis. Since the eigenvectors $\hat{Q}$ of $\hat{W}$ are orthonormal by the spectral theorem, they cannot coincide with $\hat{Q}_s$, and therefore $\hat{C} \hat{Q} \neq \tilde{B}$.

Channel 2 (P_2 corrupts the spectral decomposition). Even if $P_1 = 0$ — so that $\hat{C} = \tilde{C}$ and the whitening step is unaffected — the whitened matrix becomes

$$\hat{W}|_{P_1=0} = \tilde{C}^{-1} \tilde{\Sigma}^{(2)} \tilde{C}^{-T} + \tilde{C}^{-1} P_2 \tilde{C}^{-T} = \tilde{W} + \tilde{C}^{-1} P_2 \tilde{C}^{-T}, \quad (2.25)$$

which is a perturbation of the correctly specified $\tilde{W}$. Its eigenvectors are therefore those of the perturbed matrix, not of $\tilde{W}$, and do not coincide with the columns of $\tilde{C}^{-1}\tilde{B}$.

The two channels compound one another in general. Consequently, the recovered impact matrix

$$\hat{B} = \hat{C}\hat{Q}, \quad (2.26)$$

which mirrors Step 5 of the identification algorithm but applies the contaminated Cholesky factor $\hat{C}$, produces biased estimates. Identification is generically restored when $P_1 = P_2 = 0$, i.e. in the absence of any omitted variable. We now proceed to articulate these claims into more formal propositions.

2.3 Inconsistency of the CV Identifier

The researcher applies the CV estimator to the misspecified model, computing the whitened matrix

$$\hat{W} = \hat{C}^{-1}\hat{\Sigma}^{(2)}\hat{C}^{-T} \quad (2.27)$$

and extracting its eigenvectors $\hat{Q} = [\hat{q}_1, \hat{q}_2, \hat{q}_3]$; from these, the structural impact matrix is recovered using the contaminated back-transformation $\hat{B} = \hat{C}\hat{Q}$, in agreement with Step 5 of the identification algorithm. The whitened matrix for the correctly specified 3×3 system would instead be

$$\tilde{W} = \tilde{C}^{-1}\tilde{\Sigma}^{(2)}\tilde{C}^{-T}, \quad (2.28)$$

with eigenvectors $\tilde{Q} = [\tilde{q}_1, \tilde{q}_2, \tilde{q}_3]$ and structural impact matrix $\tilde{B} = \tilde{C}\tilde{Q}$. Define

$$E_W = \hat{W} - \tilde{W}. \quad (2.29)$$

Proposition 2.1 (Inconsistency under omitted variable). *Let $\hat{\Sigma}^{(k)} = \tilde{\Sigma}^{(k)} + P_k$ for $k = 1, 2$, where $\tilde{\Sigma}^{(k)}$ are the structural covariance matrices and P_k are the contaminations due to an omitted variable. Let $\tilde{\Sigma}^{(1)} = \tilde{C}\tilde{C}^T$ be the true Cholesky factorisation and assume that the eigenvalues of $\tilde{W}$ are distinct. Then, generically, $\hat{B} \neq \tilde{B}$, so that the CV estimator is inconsistent.*

Proof. Suppose for contradiction that $\hat{B} = \tilde{B}$. Since $\hat{B} = \hat{C}\hat{Q}$, this implies $\hat{C}\hat{C}^T = \tilde{C}\tilde{C}^T$, i.e. $\hat{\Sigma}^{(1)} = \tilde{\Sigma}^{(1)}$, and therefore $P_1 = 0$. With $P_1 = 0$ we have $\hat{C} = \tilde{C}$, so $\hat{B} = \tilde{B}$ further implies $\hat{Q} = \tilde{Q}$, and hence $\hat{\Sigma}^{(2)} = \hat{C}\hat{Q}\hat{\Lambda}^{(2)}\hat{Q}^T\hat{C}^T = \tilde{B}\hat{\Lambda}^{(2)}\tilde{B}^T = \tilde{\Sigma}^{(2)}$, giving $P_2 = 0$. Thus $\hat{B} = \tilde{B}$

requires $P_1 = P_2 = 0$, i.e. the complete absence of any omitted variable. Since $P_k \succ 0$ under Assumption A.1, this is a contradiction. $\square$

Proposition 2.2 (Decomposition of the identification error). *The inconsistency arises from two distinct perturbation channels: (i) the distortion of the Cholesky factor induced by P_1 , and (ii) the perturbation of the eigenvectors of the whitened matrix induced by P_2 . The resulting bias in the structural impact matrix admits the exact decomposition*

$$\hat{B} - \tilde{B} = \underbrace{(\hat{C} - \tilde{C})\tilde{Q}}_{\text{whitening error}} + \underbrace{\tilde{C}(\hat{Q} - \tilde{Q})}_{\text{eigenvector error}} + \underbrace{(\hat{C} - \tilde{C})(\hat{Q} - \tilde{Q})}_{\text{interaction term}}. \quad (2.30)$$

This decomposition highlights that misspecification affects both stages of the CV identification algorithm simultaneously. The first term reflects the bias in the back-transformation step, the second reflects the rotation of structural directions in the whitened space, and the third captures the non-linear interaction between the two errors, which becomes non-negligible when both P_1 and P_2 are large. An obvious implication for the researcher would be to perform the whitening within the less perturbed period, in order to decrease as much as possible the whitening error and the interaction term.

Proof. The decomposition follows from a straightforward algebraic identity. Since $\hat{B} = \hat{C}\hat{Q}$ and $\tilde{B} = \tilde{C}\tilde{Q}$,

$$\begin{aligned} \hat{B} - \tilde{B} &= \hat{C}\hat{Q} - \tilde{C}\tilde{Q} \\ &= \hat{C}\hat{Q} - \tilde{C}\hat{Q} + \tilde{C}\hat{Q} - \tilde{C}\tilde{Q} \\ &= (\hat{C} - \tilde{C})\hat{Q} + \tilde{C}(\hat{Q} - \tilde{Q}). \end{aligned} \quad (2.31)$$

Writing $\hat{Q} = \tilde{Q} + (\hat{Q} - \tilde{Q})$ in the first term and expanding yields the claimed decomposition. The three terms correspond precisely to Channel 1, Channel 2, and their interaction. $\square$

Conditions for the individual channels to vanish The decomposition reveals that $\hat{B} = \tilde{B}$ requires both $\hat{C} = \tilde{C}$ and $\hat{Q} = \tilde{Q}$ simultaneously, which by the proof of Proposition 2.1 is equivalent to $P_1 = P_2 = 0$. It is nonetheless instructive to ask under which - weaker - conditions each channel vanishes independently, if it ever does so.

Channel 1 vanishes if and only if $\hat{C} = \tilde{C}$, which holds if and only if $P_1 = 0$: the regime-1 contamination must be exactly zero.

Channel 2 vanishes if and only if $\hat{Q} = \tilde{Q}$, i.e. the eigenvectors of $\hat{W} = \tilde{W} + E_W$ coincide with those of $\tilde{W}$. By the spectral theorem, this holds if and only if E_W and $\tilde{W}$ share the same eigenbasis, which is equivalent to the commutativity condition⁷

$$[E_W, \tilde{W}] = E_W \tilde{W} - \tilde{W} E_W = 0. \quad (2.32)$$

In terms of the eigenbasis $\{\tilde{q}_j\}$ of $\tilde{W}$, this condition requires that all off-diagonal components of E_W vanish:

$$\tilde{q}_i^T E_W \tilde{q}_j = 0 \quad \text{for all } i \neq j. \quad (2.33)$$

This has a direct geometric interpretation, fully consistent with the discussion of Section 2.4.1 (the reader might want to skip this paragraph and come back after reading it): it is precisely the off-diagonal components of E_W in the eigenbasis that rotate the principal axes of the whitened ellipsoid, as shown in equation (2.40). When (2.33) holds, the perturbation acts purely along the existing directions in the whitened space — scaling the semi-axes of the ellipsoid without rotating them — so the eigenvectors are unaffected. Economically, this means that the omitted variable contamination must not introduce covariance *between* whitened structural directions in the second regime; it may only alter the variances along each direction separately: the system misinterprets the perturbation as a change in magnitude of a shock it already knows, rather than as a new spurious direction. This does not imply a correct retrieval of the structural impact vectors, as the back-transformation via the contaminated Cholesky introduces a new source of distortion.

2.4 A Davis–Kahan Bound on the Identification Error

Proposition 2.1 establishes inconsistency qualitatively. We now want to find a bound to the bias introduced by the omission. Both $\tilde{W}$ and $\hat{W}$ are symmetric positive definite, so their eigenvectors are orthonormal by the spectral theorem and Davis-Kahan can be applied directly *in the whitened space*. The bound is conservative in nature, because it accounts for a worst-case scenario where the perturbation impacts the system in the most vulnerable direction. It is, nonetheless, informative with regard to the determinants of the bias induced by the perturbation: nominally, the spectral gap and the magnitude of the perturbation itself. In principle, it might also be used as a *parsimony filter*: in that case, it would quantify how large the omission-induced perturbation E_W would need to be to materially distort identification. To be precise, then, while Proposition 2.1 establishes inconsistency

⁷ See Lemma A.3 in the Appendix

qualitatively, the Davis–Kahan bound translates this into a structural characterisation: it reveals that the identification error is governed by two quantities - the magnitude of the perturbation $\|E_W\|_{\text{op}}$ and the spectral gap δ_j - rather than providing a tight empirical measure of the bias.

Lemma 2.3 (Davis–Kahan bound). *Under the assumptions of Proposition 2.1, let*

$$\delta_j = \min(\mu_{j-1} - \mu_j, \mu_j - \mu_{j+1}) \quad (2.34)$$

be the spectral gap of $\tilde{W}$ at position j , where $\mu_1 \geq \mu_2 \geq \mu_3$ are the eigenvalues of $\tilde{W}$ ⁸ under correct specification, with the conventions $\mu_0 = +\infty$ and $\mu_4 = -\infty$. Assume $\delta_j > 0$, which guarantees that μ_j is a simple eigenvalue and $\tilde{q}_j$ is uniquely defined up to sign. Then

$$\sin \Theta(\hat{q}_j, \tilde{q}_j) \leq \frac{2\|E_W\|_{\text{op}}}{\delta_j}, \quad (2.35)$$

and there exists a sign $o_j \in \{-1, +1\}$ such that

$$\|o_j \hat{b}_j - \tilde{b}_j\| \leq \|\hat{C} - \tilde{C}\|_{\text{op}} + \frac{2^{3/2} \lambda_{\max}(\hat{\Sigma}^{(1)})^{1/2} \|E_W\|_{\text{op}}}{\delta_j}, \quad (2.36)$$

where $\hat{b}_j = \hat{C}\hat{q}_j$ is the estimated structural vector from Step 5 of the identification algorithm.

Proof. $\tilde{W}$, $\hat{W}$, and $E_W = \hat{W} - \tilde{W}$ are all symmetric matrices. Applying Theorem 2 of Yu, Wang, and Samworth (2015) with $r = s = j$ yields (2.35) directly.

Since in our case of interest the eigenspaces are one-dimensional, the orthogonal matrix appearing in Theorem 2 of Yu, Wang, and Samworth (2015) reduces to a sign $o_j \in \{-1, +1\}$, and the second bound of that theorem gives directly

$$\|o_j \hat{q}_j - \tilde{q}_j\|_2 \leq \frac{2^{3/2} \|E_W\|_{\text{op}}}{\delta_j}. \quad (*)$$

For the structural-space bound, recall that $\hat{b}_j = \hat{C}\hat{q}_j$ and $\tilde{b}_j = \tilde{C}\tilde{q}_j$. Adding and subtracting $\tilde{C}(o_j \hat{q}_j)$ gives

$$o_j \hat{b}_j - \tilde{b}_j = (\hat{C} - \tilde{C})(o_j \hat{q}_j) + \tilde{C}(o_j \hat{q}_j - \tilde{q}_j).$$

Taking Euclidean norms and applying the triangle inequality,

$$\|o_j \hat{b}_j - \tilde{b}_j\|_2 \leq \underbrace{\|(\hat{C} - \tilde{C})(o_j \hat{q}_j)\|_2}_{\text{Cholesky error}} + \underbrace{\|\tilde{C}(o_j \hat{q}_j - \tilde{q}_j)\|_2}_{\text{Davis–Kahan}}.$$

⁸ Note that $\mu_j = \lambda_j^{(2)}$, the structural volatility ratio of shock j in regime 2, as defined in equation (2.2).

For the first term, since $\|o_j \hat{q}_j\|_2 = 1$,

$$\|(\hat{C} - \tilde{C})(o_j \hat{q}_j)\|_2 \leq \|\hat{C} - \tilde{C}\|_{\text{op}}.$$

For the second term,

$$\|\tilde{C}(o_j \hat{q}_j - \tilde{q}_j)\|_2 \leq \|\tilde{C}\|_{\text{op}} \|o_j \hat{q}_j - \tilde{q}_j\|_2.$$

The singular values of the Cholesky factor $\tilde{C}$ are the square roots of the eigenvalues of $\tilde{C}^T \tilde{C}$, which coincide with those of $\tilde{C} \tilde{C}^T = \tilde{\Sigma}^{(1)}$ by the identity $\text{sp}(AB) = \text{sp}(BA)$ for square matrices. Hence $\|\tilde{C}\|_{\text{op}} = \lambda_{\max}(\tilde{\Sigma}^{(1)})^{1/2}$. Since $P_1 = \hat{\Sigma}^{(1)} - \tilde{\Sigma}^{(1)}$ is positive semidefinite, Weyl's inequality (cf. next subsection) gives $\lambda_{\max}(\tilde{\Sigma}^{(1)}) \leq \lambda_{\max}(\hat{\Sigma}^{(1)})$, so that

$$\|\tilde{C}\|_{\text{op}} \leq \lambda_{\max}(\hat{\Sigma}^{(1)})^{1/2}.$$

Combining this with (*) and the bounds on both terms gives (2.36). $\square$

Remark 2.4 (Decomposition of the structural bias). Bound (2.36) makes explicit the two channels of Remark 2.2: the term $\|\hat{C} - \tilde{C}\|_{\text{op}}$ corresponds to the Cholesky error (Channel 1), while the term with δ_j in the denominator corresponds to the eigenvector rotation in the whitened space (Channel 2). Davis–Kahan controls only Channel 2, as it applies in the whitened space; Channel 1 must be bounded separately via the contamination P_1 of the Cholesky factor.

2.4.1 Geometric Interpretation of the Davis–Kahan Bound

The bound derived in Lemma 2.3 has a clean geometric reading. We now develop it in some detail. The argument proceeds in logical steps: first, we explain what the whitened matrix $\tilde{W}$ represents as a geometric object; second, we interpret the spectral gap δ_j as a measure of the *eccentricity* of that object; last, we explain how the perturbation E_W rotates the principal axes and why the gap determines the sensitivity of that rotation.

The Whitened Matrix as an Ellipsoid

Section 2.1 established that the whitening transformation maps the identification problem, via a change of basis, into a space where the regime-1 covariance becomes the identity. In this space, the cloud of whitened residuals in regime 1 is a sphere - all directions are equally

variable⁹. The whitened matrix

$$\tilde{W} = \tilde{C}^{-1} \tilde{\Sigma}^{(2)} \tilde{C}^{-T} \quad (2.37)$$

then expresses how that sphere *deforms* in regime 2: some directions become more variable (longer semi-axes), others less (shorter semi-axes).

Indeed, $\tilde{W}$ is a symmetric positive definite matrix, and by the spectral theorem it admits the eigendecomposition $\tilde{W} = \tilde{Q} \text{diag}(\mu_1, \dots, \mu_n) \tilde{Q}^T$, where $\mu_1 > \dots > \mu_n > 0$ are the eigenvalues and $\tilde{Q} = [\tilde{q}_1, \dots, \tilde{q}_n]$ is an orthonormal matrix of eigenvectors. The one-standard-deviation "boundary" of a distribution with covariance $\tilde{W}$ is an ellipsoid with principal axes $\tilde{q}_j$ and semi-radii $\sqrt{\mu_j}$.

Two geometric features of this ellipsoid are immediately readable from $\tilde{W}$:

- *Orientation*: the eigenvectors $\tilde{q}_j$ point along the principal axes. These are precisely the structural directions we seek to identify - back-transforming through $\tilde{C}$ gives the structural impact vectors $\tilde{b}_j = \tilde{C} \tilde{q}_j$.
- *Shape*: the eigenvalues $\mu_j = \lambda_j^{(2)}$ determine the relative lengths of the axes. In the normalisation $\Lambda^{(1)} = I$, these are the volatility ratios of the structural shocks between regime 2 and regime 1. A shock whose variance doubles in regime 2 has $\mu_j = 2$ and thus a semi-axis of length $\sqrt{2}$. A shock with unchanged variance has $\mu_j = 1$ and thus a unit semi-axis.

The identification procedure of Section 2.1 is, then, geometrically equivalent to *finding the semi-axes of the ellipsoid*. Identification fails if two semi-axes are equal — in that case, the corresponding section of the ellipsoid is a circle, with *infinitely many* equally valid pairs of orthogonal axes, not just one.

Now, in Figure 1, we provide a graphical illustration, inspired by the discussion in Rigobon (2003). It sheds light on the subsequent steps of the identification procedure. The data comes from a simulated, bivariate DGP with $\lambda_1^{(2)} = 3$, $\lambda_2^{(2)} = 0.8$, $T = 1000$ observations per regime. *Left*: the original residual space (u_{1t}, u_{2t}) . The ellipses of $\tilde{\Sigma}_1$ and $\tilde{\Sigma}_2$ are both tilted and overlapping; structural directions are not directly discernible. *Centre*: the whitened space $(u_{1t}^w, u_{2t}^w) = C^{-1}u_t$. The Cholesky transformation maps Σ_1 to the unit sphere ($W_1 = I$ by construction); $W_2 = C^{-1}\tilde{\Sigma}_2 C^{-T}$ is the identifying ellipse, with arrows indicating its principal axes $\tilde{q}_j$. The ellipse is, however, still tilted relative to the coordinate axes. *Right*: eigenbasis

⁹ For the intuition behind this, a nice visualization can be found at <https://pantelis.github.io/cs677/docs/common/lectures/optimization/whitening/>.

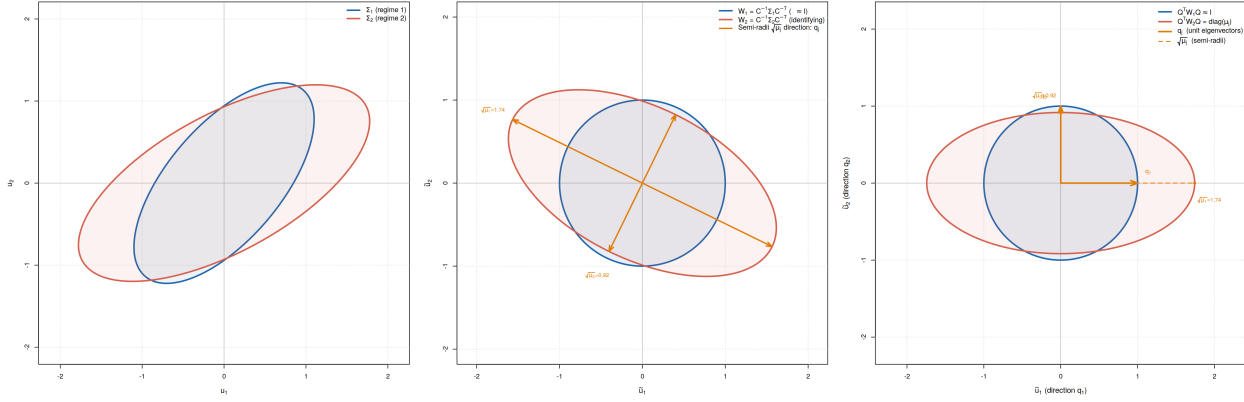


Fig. 1: Geometry of the whitening procedure in three steps

of W_2 , obtained by the rotation $u^w \mapsto Qu^w$. The ellipse is now aligned with the coordinate axes. The semi-radii $\sqrt{\mu_j}$ are the volatility ratios of the structural shocks in regime 2. The coordinate axes are the eigenvectors $\tilde{q}_j$ of the whitened matrix. The structural impact vectors are recovered as $\tilde{b}_j = C\tilde{q}_j$, which live in the original space of y_t .

The Spectral Gap as Eccentricity

The spectral gap

$$\delta_j = \min(\mu_{j-1} - \mu_j, \mu_j - \mu_{j+1}) \quad (2.38)$$

measures the difference in length between the j -th semi-axis and its nearest neighbours. Geometrically, δ_j is a measure of the *local eccentricity*¹⁰ of the ellipsoid in the neighbourhood of axis j : how different the ellipsoid looks from a sphere when viewed in the plane spanned by $\tilde{q}_j$ and its adjacent directions. To see why this matters for identification, consider the two-dimensional case with $n = 2$, $\mu_1 > \mu_2$. The ellipsoid is an ellipse with semi-major axis $\sqrt{\mu_1}$ and semi-minor axis $\sqrt{\mu_2}$.

Large gap ($\delta_j = \mu_1 - \mu_2$ large). The ellipse is strongly eccentric: its major axis is much longer than the minor one. Visually, the orientation of the major axis is unambiguous — it points clearly in one direction. Any perturbation of the covariance matrix that slightly changes

¹⁰ This visualization is valid only for relatively small values of the eigenvalues - coherently with their interpretation as volatility breaks -, as the spectral gap is properly defined as their absolute difference, not as their ratio. That said, for most macroeconomic SVAR settings, the small absolute values of μ_j do not invalidate the eccentricity interpretation.

the shape of the ellipse will not substantially alter where the major axis points, because the elongated shape provides a strong geometric signal.

Small gap ($\delta_j \approx 0$). When both eigenvalues coincide, the ellipse degenerates into a circle. Every direction in the plane is now an eigenvector with the same eigenvalue; the consequence is that no axis is geometrically distinguishable from any other. There is nothing to “resist” a rotation: an arbitrarily small perturbation E_W is sufficient to break the symmetry and select a new pair of axes. This new pair may be oriented completely differently from the nominal one. More technically, the eigenspace is no longer a pair of isolated rays but an entire two-dimensional subspace, so the eigenvectors are undefined up to an arbitrary rotation within that subspace. The Davis–Kahan bound reflects just this: as $\text{gap} \rightarrow 0$, the upper bound on $\sin \theta$ diverges, signalling that no stability is guaranteed.

This is the precise geometric reasoning behind the weak-identification results described in Lewis (2022): identification via heteroskedasticity is particularly fragile when the volatility ratios $\lambda_j^{(2)}$ of different shocks are nearly equal. When $\lambda_j^{(2)} \approx \lambda_i^{(2)}$, the corresponding section of the whitened ellipsoid is nearly circular, and the two structural directions cannot be reliably distinguished from second-moment information.

How the Perturbation Rotates the Axes

When the researcher works with the contaminated matrix $\hat{W} = \tilde{W} + E_W$ instead of the true $\tilde{W}$, the ellipsoid is *deformed*: its shape and orientation both change as a function of P_1 and P_2 . The operator norm $\|E_W\|_{\text{op}}$ is the magnitude of the largest deformation — in fact, it is the maximum over all unit vectors u of $|u^T E_W u|$, i.e. the largest possible change in any single quadratic form. This is the radius of the “ball of distortion” that E_W can impose on the ellipsoid.

The deformation has two effects, which are governed by different theorems.

1. *Shifting the semi-radii* (Weyl’s inequality): Weyl’s theorem (Weyl, 1912) states that the spectrum of Hermitian matrices is stable under perturbation. In our case, we have $|\hat{\mu}_j - \mu_j| \leq \|E_W\|_{\text{op}}$. Applying the mean value theorem (Bottazzi, 2023) to $f(x) = \sqrt{x}$ yields $|\sqrt{\hat{\mu}_j} - \sqrt{\mu_j}| \lesssim \|E_W\|_{\text{op}} / (2\sqrt{\mu_j})$ for small perturbations. Eigenvalues are therefore stable: the lengths of the axes change by at most $\|E_W\|_{\text{op}}$.
2. *Rotating the axes* (Davis–Kahan): eigenvectors can be highly unstable. The rotation

of axis j depends critically on the spectral gap.

To understand the rotation mechanically, decompose E_W in the eigenbasis of $\tilde{W}$ (Lang, 2014). Write

$$(E_W)_{ij} = \tilde{q}_i^T E_W \tilde{q}_j. \quad (2.39)$$

The *diagonal* components $(E_W)_{jj}$ simply add to the eigenvalues: they shift the semi-radii but do not move the axes. The *off-diagonal* components $(E_W)_{ij}$ with $i \neq j$ are the dangerous ones: they *mix* eigenvectors i and j , pulling the eigenvectors of $\hat{W}$ away from those of $\tilde{W}$. The smaller the gap between μ_i and μ_j , the larger the rotation induced by any given off-diagonal entry.

In the two-dimensional case, the exact rotation angle of the first eigenvector under a perturbation with off-diagonal component $e_{12} = \tilde{q}_1^T E_W \tilde{q}_2$ is given by¹¹

$$\theta = \frac{1}{2} \arctan \frac{2|e_{12}|}{\delta_j}. \quad (2.40)$$

This formula makes the geometry transparent. The rotation angle is determined by the ratio of two quantities:

- $|e_{12}|$ is the *mixing strength*: how much of E_W is directed toward blending the two structural directions. This depends on the direction of E_W relative to the eigenbasis, not just its magnitude. A perturbation that is diagonal in the eigenbasis has $e_{12} = 0$ and produces no rotation at all, regardless of $\|E_W\|_{\text{op}}$.
- $\delta_j = \mu_1 - \mu_2$ is the *restoring force*: it resists the mixing. The smaller the gap, the easier it is to "confuse" the two axes, because any rotation would produce an ellipse with markedly different semi-radii from $\tilde{W}$.

The Davis–Kahan theorem bounds the rotation over all possible directions of E_W . Since $|e_{12}| \leq \|E_W\|_{\text{op}}$ (the off-diagonal component cannot exceed the operator norm), equation (2.40) implies¹²

$$\sin \Theta(\hat{q}_j, \tilde{q}_j) \leq \frac{\|E_W\|_{\text{op}}}{\delta_j}. \quad (2.41)$$

¹¹ This follows from the eigenvalue problem for $\tilde{W} + E_W = \text{diag}(\mu_1, \mu_2) + \begin{bmatrix} (E_W)_{11} & e_{12} \\ e_{12} & (E_W)_{22} \end{bmatrix}$ in the eigenbasis. The rotation angle is $\frac{1}{2} \arctan(2e_{12}/(\mu_1 - \mu_2 + (E_W)_{11} - (E_W)_{22}))$, which simplifies to (2.40) for a purely off-diagonal perturbation. The formula for $\tan(2\theta)$ is obtained by imposing that $R^T M R$ be diagonal for a rotation matrix R , which yields $\tan(2\theta) = 2b/(a - c)$; see, e.g., Golub (1996).

¹² Bounding $\sin(\alpha) \leq \alpha$ and $\arctan(x) \leq x$ gives $\sin \theta \leq \frac{1}{2} \arctan(2\|E\|/\delta) \leq \|E\|/\delta$.

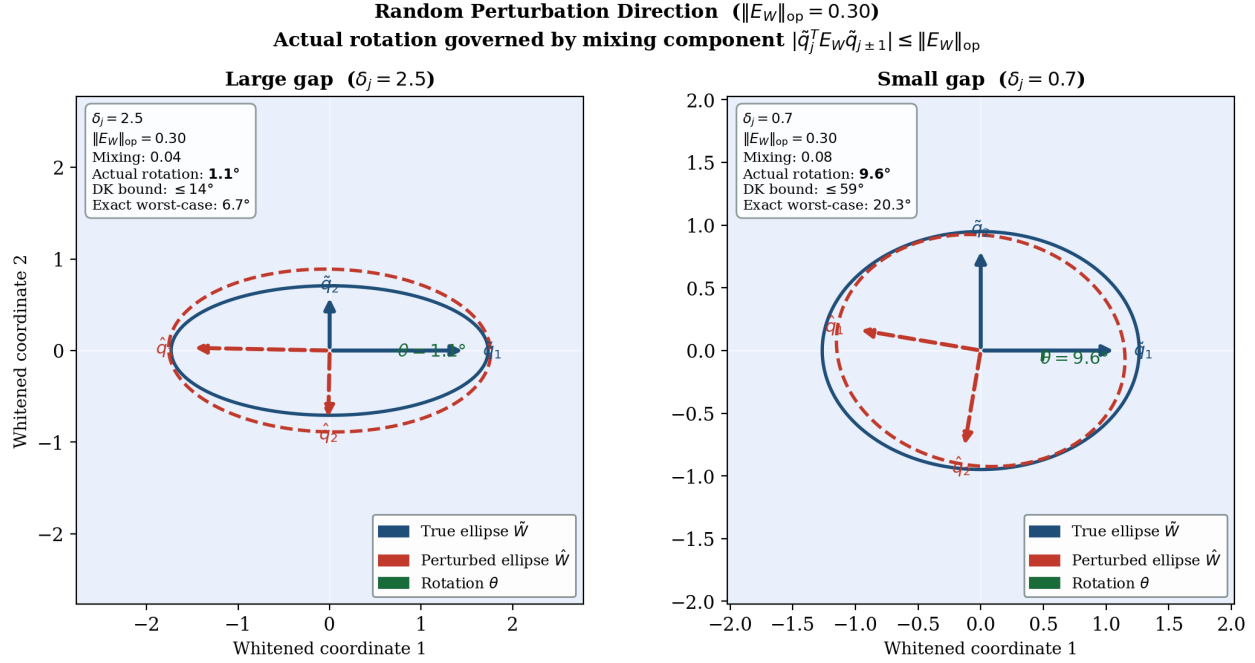


Fig. 2: Random perturbation direction, $\|E_W\|_{\text{op}} = 0.30$.

In fact, the exact worst-case rotation – achieved when E_W is entirely off-diagonal in the eigenbasis, $|e_{12}| = \|E_W\|_{\text{op}}$ – satisfies

$$\theta_{\text{worst}} = \frac{1}{2} \arctan \frac{2 \|E_W\|_{\text{op}}}{\delta_j} \approx \frac{\|E_W\|_{\text{op}}}{\delta_j} \quad \text{for small ratios,} \quad (2.42)$$

which is approximately *half* of the bound found in Yu, Wang, and Samworth (2015). This latter bound is therefore conservative by at least a factor of 2 even in the worst case, and can be far more conservative when the perturbation is not directed toward maximal mixing, which unfortunately limits its empirical relevance for diagnostic purposes. It nevertheless remains a useful heuristic tool for understanding the dynamics of a perturbation due to the omission of a relevant variable. The example that follows better illustrates these considerations.

Illustration

Figures 2 and 3 illustrate the geometry for a two-dimensional system with eigenvalues $\mu_1 \in \{3.0, 1.6\}$ and $\mu_2 \in \{0.5, 0.9\}$, giving a large gap ($\delta_j = 2.5$) in the left panel and a small gap ($\delta_j = 0.7$) in the right panel. In both figures, $\|E_W\|_{\text{op}} = 0.30$. The true ellipse (solid blue) and the perturbed ellipse (dashed orange) are shown with their respective principal axes. The arc indicates the actual rotation θ of the first eigenvector.

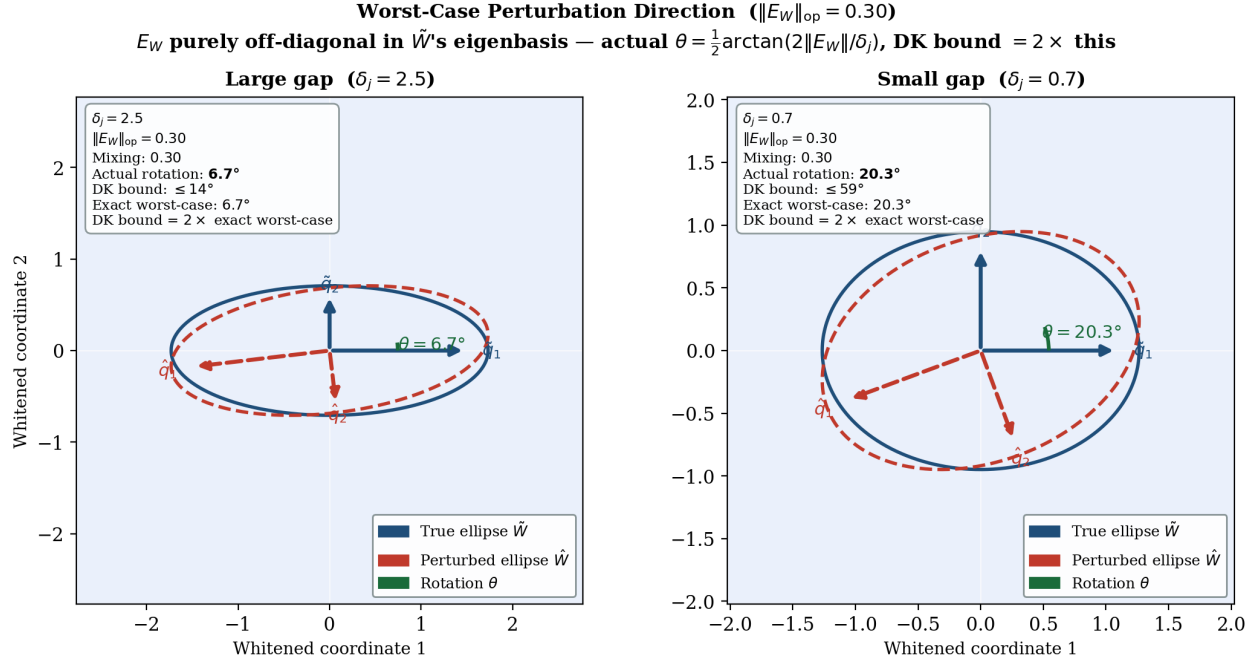


Fig. 3: Worst-case perturbation direction: E_W is entirely off-diagonal in the eigenbasis of $\tilde{W}$, so $|e_{12}| = \|E_W\|_{\text{op}} = 0.30$.

Figure 2 uses a random perturbation direction. The box in each panel reports the mixing component $|e_{12}|$, which is strictly less than $\|E_W\|_{\text{op}}$ because the random E_W is not entirely off-diagonal in $\tilde{W}$'s eigenbasis. The actual rotation is therefore well below the DK bound in both panels, illustrating that the bound's conservativeness is generic, not exceptional.

Figure 3 uses the worst-case direction: E_W is entirely off-diagonal in the eigenbasis, so $|e_{12}| = \|E_W\|_{\text{op}} = 0.30$. Now the actual rotation exactly matches θ_{worst} from (2.42) – approximately half the DK bound. The contrast between the two panels is now fully determined by the spectral gap: the large-gap ellipsoid (left) is eccentric and resists the perturbation; the small-gap ellipsoid (right) is nearly circular and amplifies it. The DK bound correctly orders the two cases but overstates both by the usual factor of 2.

3 Monte Carlo Evidence

The theoretical results of Section 2 yield a set of empirically testable predictions. Proposition 2.1 establishes that the CV estimator, when applied to a misspecified model, does not converge to $\tilde{B}$ but to a distinct target $B^* \neq \tilde{B}$, whereas under irrelevant omission - when stands $P_1 \approx P_2 \approx 0$ - consistency should be restored. Lemma 2.3 further predicts that the angular bias $\sin \Theta(\hat{q}_j, \tilde{q}_j)$ is bounded by $2\|E_W\|_{\text{op}}/\delta_j$: holding the spectral gap fixed, the bound increases with the perturbation norm $\|E_W\|_{\text{op}}$; additionally, holding the perturbation norm fixed, it decreases with the gap δ_j . Furthermore, Proposition 2.2 shows that the identification error decomposes into two contributions of qualitatively different source: a Cholesky distortion, driven by the regime-1 contamination P_1 and which does not depend on the variance of the omitted shock in regime 2, and an eigenvector rotation driven by the perturbation E_W , which in turn does. The Monte Carlo experiments described in this section aim to test each one of these predictions.

3.1 Simulation Design

Data-Generating Process

The DGP is a VAR(2) with $n = 4$ variables, $y_t = A_1 y_{t-1} + A_2 y_{t-2} + B w_t$, where the coefficient matrices and the structural impact matrix are:

$$A_1 = \begin{pmatrix} 0.40 & 0.08 & -0.05 & 0.10 \\ 0.05 & 0.45 & 0.08 & 0.07 \\ 0.00 & 0.15 & 0.50 & 0.05 \\ 0.10 & 0.05 & 0.05 & 0.50 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0.05 & 0.00 & 0.00 & 0.02 \\ 0.00 & 0.05 & 0.00 & 0.02 \\ 0.00 & 0.00 & 0.05 & 0.01 \\ 0.02 & 0.01 & 0.01 & 0.10 \end{pmatrix}, \quad (3.1)$$

$$B = \begin{pmatrix} 1.3 & 1.0 & -0.7 & 0.8 \\ 0.9 & -0.8 & -0.6 & 0.7 \\ 0.7 & 0.6 & 1.3 & 0.6 \\ -0.6 & 0.7 & -0.5 & 1.5 \end{pmatrix}. \quad (3.2)$$

All eigenvalues of the VAR(2) companion matrix lie strictly inside the unit circle, so that the process is stable and covariance-stationary. The autoregressive matrices A_1 and A_2 are deliberately chosen to be non-diagonal, creating realistic cross-variable lag dynamics. Notably, the two columns $a_i = A_i[1 : 3, 4]$ associated with the omitted variable are non-zero for both lags, so that y_{4t} influences $\tilde{y}_t$ with non-trivial dynamics. The fourth column

$c = B[1:3, 4] = (0.8, 0.7, 0.6)^T$ of the structural impact matrix is likewise non-negligible, so both channels of Proposition 2.2 are active.

The structural shocks w_t follow the two-regime heteroskedastic specification of Section 2.1, with $\Lambda^{(1)} = I_4$ in regime 1. Regime-2 variances are parameterised as

$$\lambda_j^{(2)} = (1 - \lambda) + \lambda d_j, \quad (3.3)$$

where $\lambda = 0.8$ is a fixed mixing coefficient, and d_j determines the target variance level in regime 2. For the three observed shocks, $d_1 = 2$, $d_2 = 3$, $d_3 = 4$, giving $\lambda_1^{(2)} = 1.8$, $\lambda_2^{(2)} = 2.6$, $\lambda_3^{(2)} = 3.4$. The fourth shock has $d_4 = \lambda_4$, a free parameter varied across experiments. Since the eigenvalues of the correctly specified whitened matrix $\tilde{W}$ coincide with the structural volatility ratios $\lambda_j^{(2)}$ (cf. the footnote in Lemma 2.3), the spectral gaps of the three-variable benchmark are

$$\delta_j = \lambda \cdot (d_{j+1} - d_j) = 0.8 \quad (3.4)$$

for all adjacent pairs j under the baseline parameterisation.

Each simulation generates $T_{\text{tot}} = T + 200$ observations, of which the first 200 serve as a burn-in period aimed at removing the influence of arbitrary initial conditions; hence, they are discarded prior to estimation (a common practice in simulation studies: see Herwartz, Lange, and Maxand (2019), who run MC with a similar approach). The structural break is then placed at the midpoint of the generated sample, so that approximately $T/2$ observations belong to each regime and the samples are balanced. In all experiments, the break location is treated as known to the researcher: indeed, this eliminates break-detection uncertainty as a confounding factor, and isolates the pure effect of misspecification on identification.

Omission Scenarios

Two scenarios are maintained across simulations, and provide the basis for comparison in every figure.

Relevant omission. The structural impact matrix B and the autoregressive matrices A_1 , A_2 are used at their full values, as specified in (3.1) and (3.2). The fourth variable contributes to both the contemporaneous covariance structure and the dynamics of $\tilde{y}_t$. Under this scenario, the regime- k contaminations $P_k = \mathbb{E}[\eta_t \eta_t^T | k]$ are strictly positive definite if the Assumption A.1 in the Appendix applies, and Proposition 2.1 declares that the CV estimator is inconsistent.

Irrelevant omission. The fourth column of B and the sub-matrices $A_i[1:3, 4]$ are attenuated by a factor of 0.1. The structural impact of y_{4t} on the observed system thus becomes negligible: the contaminations satisfy $P_1 \approx P_2 \approx 0$, so the conditions of Proposition 2.1 for inconsistency are actually not met. This scenario, therefore, acts as an empirical *consistency* baseline: as $T \rightarrow \infty$, the misspecified CV estimator should behave as if the correctly specified 3×3 system was being estimated, and the Frobenius error should asymptotically decline towards zero.

The baseline number of replications is $N_{\text{rep}} = 500$ for sample sizes $T \leq 1,000$. For larger samples, where Monte Carlo variance naturally decreases, the replication count is reduced, in order to reduce computational expenses, according to this rationale

$$N_{\text{rep}}(T) = \max(30, \lfloor 500 \cdot \min(1, 1000/T) \rfloor), \quad (3.5)$$

preserving uniform Monte Carlo standard errors across sample sizes. Reported confidence intervals are 90% normal intervals based on the Monte Carlo standard error.

Column Alignment and Sign Normalisation

As discussed in Section 2.1, the structural impact matrix is identified only up to permutation and sign of its columns. To compare $\hat{B}$ and $\tilde{B}$ across Monte Carlo replications, and to avoid Frobenius bias inflation caused by misaligned impact vectors, we solve

$$\hat{\pi} = \arg \max_{\pi} \sum_{j=1}^3 \frac{|\langle \hat{b}_{\pi(j)}, \tilde{b}_j \rangle|}{\|\hat{b}_{\pi(j)}\| \|\tilde{b}_j\|}, \quad (3.6)$$

where the maximum is taken over all $3! = 6$ bijections $\pi: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$. Each element π corresponds to one of the six possible reorderings of the column indices, and $\hat{\pi}(j)$ identifies which column of $\hat{B}$ is matched to the j -th column of $\tilde{B}$. The objective sums the absolute cosine similarities between matched columns, so that $\hat{\pi}$ selects the column pairing that is globally most coherent in direction, regardless of sign. It is, essentially, a way to quickly obtain the best permutation matrix to be applied to the columns of $\hat{B}$.

Problem (3.6) is a *linear sum assignment problem* (LSAP): given the 3×3 profit matrix $S_{jl} = |\langle \hat{b}_l, \tilde{b}_j \rangle| / (\|\hat{b}_l\| \|\tilde{b}_j\|)$, find the permutation maximising $\sum_j S_{j, \hat{\pi}(j)}$. We use the Hungarian algorithm found in Papadimitriou and Steiglitz (1998), as implemented in `clue::solve_LSAP` (Hornik, 2023). After permuting the columns of $\hat{B}$ according to $\hat{\pi}$, each column is sign-normalised so that its inner product with the corresponding true column is non-negative

(Axler, 2024). The Frobenius error and the angular metrics are computed on the aligned, sign-normalised matrix. This procedure is required in order to ensure that reported errors reflect genuine identification failure, rather than mere label or sign ambiguity.

3.2 Inconsistency Under Relevant Omission

Frobenius Error

Figure 4 displays the mean Frobenius error $\mathbb{E}\|\hat{B} - \tilde{B}\|_F$ as a function of sample size T (on a logarithmic scale), separately for relevant and irrelevant omission. Three contamination heteroskedasticity levels are considered: $\lambda_4 \in \{2, 4, 8\}$, labelled low, medium, and high respectively. The dashed horizontal lines mark the theoretical asymptotic targets $\|B^* - \tilde{B}\|_F$, where B^* is the population limit of the CV estimator under relevant omission, computed from the misspecified generalised eigenvalue problem using true population covariances¹³.

Under relevant omission, the mean Frobenius error exhibits a two-phase pattern. For small T , the error declines as the sample grows: this reflects the familiar reduction in finite-sample estimation noise as more data is available. Beyond $T = 3,000$, however, the error ceases to decline. Indeed, it stabilises at a positive plateau, confirming that the CV estimator is converging to a target strictly different from $\tilde{B}$. The plateau level is monotonically increasing in λ_4 : at $\lambda_4 = 2$, it is approximately 0.4 Frobenius units; at $\lambda_4 = 8$, it rises to approximately 1.4 units. Across all three contamination levels, the simulated means at $T = 30,000$ align closely with the analytically computed values of $\|B^* - \tilde{B}\|_F$. This provides direct numerical validation of Proposition 2.1 and of the population computation underlying the theoretical plateau.

Under irrelevant omission, the picture is qualitatively different. The Frobenius error declines steadily as T increases, with no sign of stabilisation through $T = 30,000$. This is a direct confirmation of the consistency of SVAR identification in a correctly specified scenario: as the structural influence of y_{4t} on $\tilde{y}_t$ is attenuated to negligible levels, the contaminations P_k vanish. As a consequence, the asymptotic target B^* collapses to $\tilde{B}$. The ordering of the plateau levels across contamination intensities is directly interpretable through Lemma 2.3.

¹³ B^* is obtained by applying Steps 2–5 of the CV identification algorithm (Section 2.1) to the contaminated population covariance matrices $\hat{\Sigma}^{(k)} = \tilde{\Sigma}^{(k)} + P_k$, where $P_k = \mathbb{E}[w_{4t}^2 | k] \cdot cc^T$ retains only the contemporaneous contamination channel. Since all quantities are evaluated at true parameter values, B^* is deterministic and coincides with $T \rightarrow \infty \hat{B}$ under misspecification (Proposition 2.1).

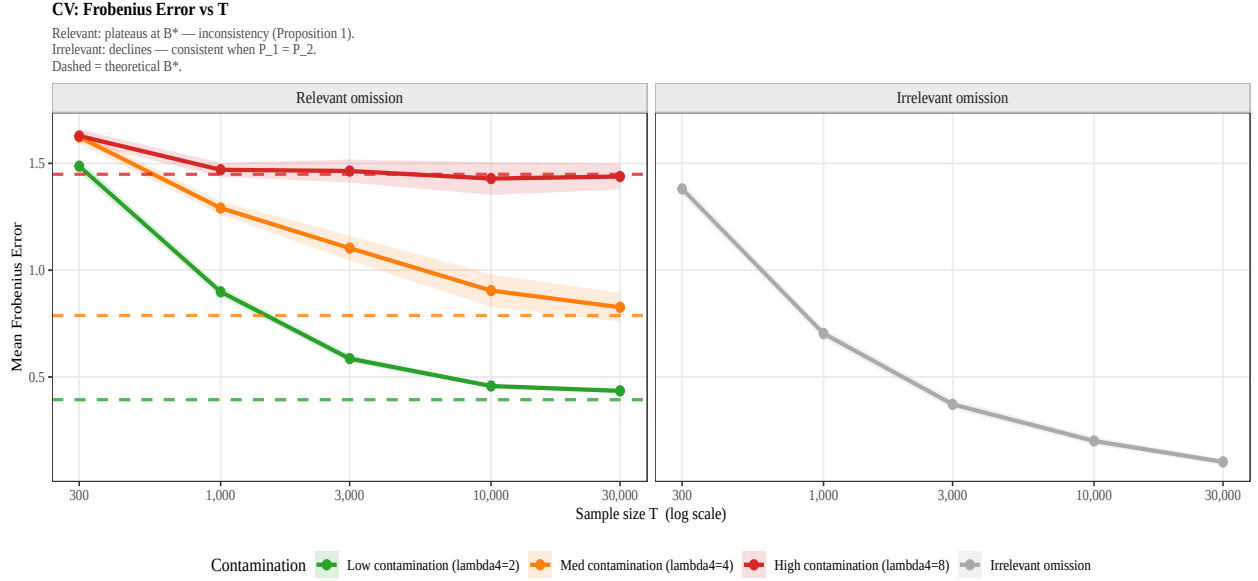


Fig. 4: Mean Frobenius error $\|\hat{B} - \tilde{B}\|_F$ as a function of sample size T under relevant and irrelevant omission.

Since the spectral gap $\delta_j = 0.8$ is held constant across contamination levels by the baseline parameterisation, the Davis–Kahan bound $2\|E_W\|_{\text{op}}/\delta_j$ increases proportionally with the perturbation norm $\|E_W\|_{\text{op}}$. A larger λ_4 implies a larger regime-2 variance for the omitted shock, a larger difference $P_2 - P_1$, and hence a larger $\|E_W\|_{\text{op}}$, all of which translate into a larger asymptotic angular rotation, as well as a higher Frobenius plateau.

Angular Bias in the Whitenesd Space

Crucially, Figure 5 complements the Frobenius analysis by reporting the mean $\sin \Theta(\hat{q}_j, \tilde{q}_j)$ directly in the whitened space. It separates between the aggregate metric (mean over $j = 1, 2, 3$, solid lines) and the most affected eigenvector j^{*14} (dashed lines, left panel only). Reporting angular distortion in the whitened space is informative for two separate reasons. First, Lemma 2.3 is stated precisely in terms of the eigenvectors of $\hat{W}$ and $\tilde{W}$, so that the whitened-space metric is actually the one directly governed by the Davis–Kahan bound. Second, the angular metric isolates the contribution of Channel 2 (eigenvector rotation) from Channel 1

¹⁴ The most-affected eigenvector j^* is identified from population quantities by applying the CV algorithm to $\hat{\Sigma}^{(1)}$ and $\hat{\Sigma}^{(2)}$ at the true parameter values, yielding contaminated eigenvectors $\hat{Q}$. We set $j^* = \arg \min_j \max_l |\hat{q}_j^T \hat{q}_l|$, i.e., the column of $\hat{Q}$ whose best match in $\hat{Q}$ is smallest – the direction most rotated by the contamination at the population level.

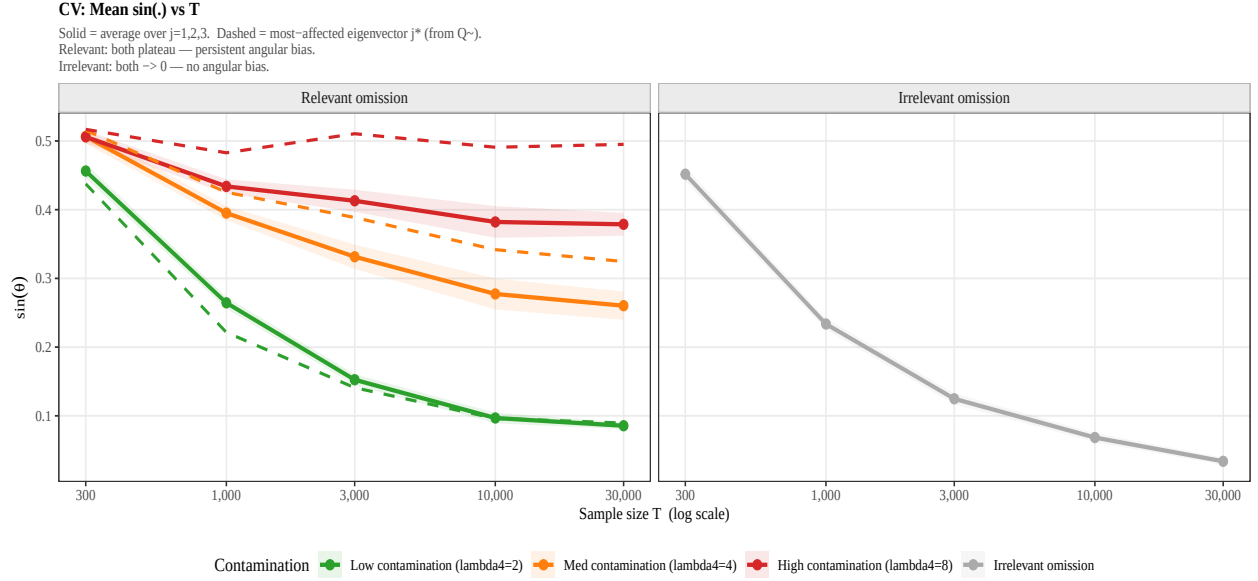


Fig. 5: Mean $\sin \Theta(\hat{q}_j, \tilde{q}_j)$ as a function of sample size T

(Cholesky distortion), since it captures only the misalignment between $\hat{Q}$ and $\tilde{Q}$ and is insensitive to the rescaling of columns by $\hat{C}$.

The qualitative pattern, in any case, mirrors Figure 4, and is coherent with our theoretical predictions: under relevant omission, both metrics plateau at positive values increasing in λ_4 ; under irrelevant omission, both converge to zero. The additional information conveyed in Figure 5 lies in the comparison between the two angular metrics. In the relevant-omission panel, the dashed curve for j^* consistently lies above the solid average curve. The perturbation concentrates its angular distortion on the structural direction most aligned with E_W , while the other eigenvectors sustain smaller rotations. This heterogeneity is precisely the mechanism that motivates tracking j^* separately, and it also has a practical implication: a researcher examining aggregate measures of identification failure risks underestimating the shock-specific exposure of the most vulnerable structural shock. Notice how the gap between the j^* metric and the average is more pronounced at higher contamination levels. This reflects the structure of the rotation formula (2.40): as λ_4 increases, $\|E_W\|_{\text{op}}$ grows, but the induced mixing components $|\tilde{q}_j^T E_W \tilde{q}_{j'}|$ are not uniformly distributed across eigenvectors. The component corresponding to j^* absorbs most of the increase, since j^* is the direction most exposed to E_W ; the remaining eigenvectors experience a comparatively smaller growth in their mixing components (and hence a smaller additional rotation). At small sample sizes, estimation noise in $\hat{Q}$ dominates the population-level rotation. It affects all three eigenvec-

tors roughly uniformly, compressing the gap between the j^* metric and the average. As T grows, the noise diminishes and the population structure of E_W emerges: j^* sustains a markedly larger rotation than the remaining eigenvectors, widening the gap between the two metrics.

The DK Numerator: Contamination Intensity

Recall that Lemma 2.3 bounds the angular bias by $2\|E_W\|_{\text{op}}/\delta_j$. Our first two simulation designs held the spectral gap fixed at $\delta_j = 0.8$ and varied λ_4 , thereby testing the numerator prediction: the identification error should increase with $\|E_W\|_{\text{op}} \propto \lambda_4$. Figure 6 provides a sharper and cleaner version of this test by plotting the *omission cost* – defined as the excess of Frobenius error attributable to the structural omission – as a function of λ_4 , for a fixed sample size $T = 10,000$.

The cost metric is defined as

$$\text{Cost}(\lambda_4) = \mathbb{E}[\|\hat{B}_{\text{rel}} - \tilde{B}\|_F] - \mathbb{E}[\|\hat{B}_{\text{irr}} - \tilde{B}\|_F], \quad (3.7)$$

where $\hat{B}_{\text{rel}}$ and $\hat{B}_{\text{irr}}$ denote the CV estimates under relevant and irrelevant omission respectively. Subtracting the irrelevant baseline removes the component of identification error due to finite-sample estimation noise that is common to both scenarios. This effectively isolates the incremental identification damage attributable solely to the structural omission. Six contamination levels are considered: $\lambda_4 \in \{1.5, 2.0, 3.0, 4.0, 6.0, 8.0\}$.

Figure 6 shows that the omission cost is strictly increasing and approximately convex in λ_4 . At $\lambda_4 = 1.5$, the cost is close to zero: the regime-2 variance of the omitted shock ($\lambda_4^{(2)} = 0.2 + 0.8 \times 1.5 = 1.4$) is only marginally larger than its regime-1 value, so the additional perturbation is small relative to the finite-sample noise at $T = 10,000$. By $\lambda_4 = 8$, the cost has risen to approximately 0.8 Frobenius units, with a tight confidence band; a confirmation that a highly volatile omitted variable can cause substantial identification failure.

The DK Denominator: Spectral Gap as Resilience

The experiments of Sections 3.2 and 3.2 tested the numerator of the Davis–Kahan bound by varying $\|E_W\|_{\text{op}}$, while holding the spectral gap fixed. The complementary prediction – that the identification error decreases with the spectral gap δ_j at constant perturbation – is now tested in Figure 7. This experiment is the Monte Carlo counterpart of the geometric

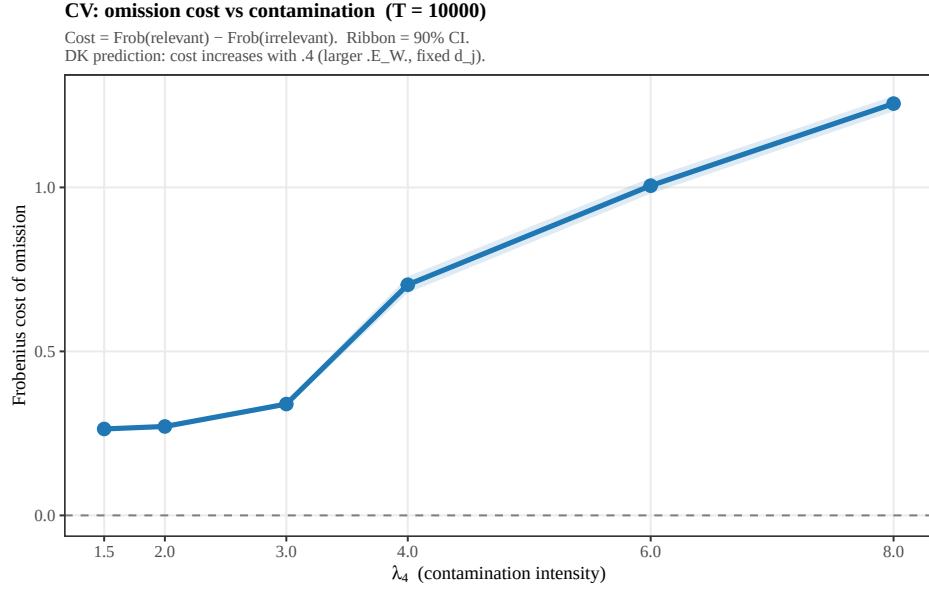


Fig. 6: Omission cost $\text{Cost}(\lambda_4)$ as a function of contamination intensity λ_4 .

discussion in Section 2.4.1, and provides direct evidence for the characterisation of δ_j as a resilience indicator.

The fundamental design challenge is to vary δ_j without simultaneously altering $\|E_W\|_{\text{op}}$. We achieve this, by introducing a separate mixing coefficient λ_{obs} that governs the regime-2 variance of the three *observed* shocks, while the omitted shock’s dynamics remain anchored to the baseline value $\lambda_{\text{fixed}} = 0.8$. Concretely, the regime-2 variance of observed shock j is $(1 - \lambda_{\text{obs}}) + \lambda_{\text{obs}} d_j$ with $d_j \in \{2, 3, 4\}$, giving spectral gaps

$$\delta_j = \lambda_{\text{obs}} \cdot (d_{j+1} - d_j) = \lambda_{\text{obs}}, \quad j = 1, 2, \quad (3.8)$$

which is linear in the experimental parameter λ_{obs} .

The perturbation E_W , on the other hand, depends only on the omitted shock’s dynamics and on the regime-1 covariance of the misspecified system. Specifically, the regime-1 contamination is $P_1 = cc^T$, where $c = B[1:3, 4]$ is the impulse vector and the omitted shock has unit variance in regime 1 by normalisation. The regime-2 contamination is $P_2 = \mu_4^{(2)} cc^T$, with $\mu_4^{(2)} = (1 - \lambda_{\text{fixed}}) + \lambda_{\text{fixed}} \lambda_4 = 1.8$ fixed by design. Since the whitening operator used to construct E_W depends only on $\hat{\Sigma}^{(1)} = \tilde{B}\tilde{B}^T + P_1$, which is independent of λ_{obs} , the perturbation magnitude $\|E_W\|_{\text{op}}$ is constant across the x -axis of Figure 7. The experiment therefore provides an exact isolation of the Davis–Kahan denominator.

Figure 7 is plotted at $T = 30,000$, a sample size at which the estimator is effectively at its asymptotic plateau for all values of λ_{obs} considered (as established by the previous asymptotic experiments). The displayed sine values, therefore, reflect genuine asymptotic angular bias. The figure displayed shows that the asymptotic angular bias for j^* is monotonically decreasing in λ_{obs} : The decline is sharp, at least up to $\lambda_{\text{obs}} = 0.5$, and has a roughly hyperbolic shape, consistent with the $1/\delta_j$ functional form of Lemma 2.3.

This result confirms that the spectral gap is not simply a technical condition for the identifiability of $\tilde{q}_j$ (which requires only $\delta_j > 0$), but crucially a quantitative measure of the structural resilience of the CV identification procedure to omission-induced perturbations. A system in which the structural shocks differ widely in their volatility ratios across regimes – a wide spectral gap – is substantially more robust to the contamination induced by an omitted variable than a system where the volatility ratios are close together.

From a practical standpoint, this finding carries an important implication for applied researchers. The robustness of the recovered impact matrix depends not only on the aggregate strength of the heteroskedasticity in the data (which determines the general signal-to-noise ratio for CV identification) but also on the *diversity* of the volatility changes across shocks. Two datasets may exhibit the same overall degree of heteroskedasticity, and yet differ substantially in their resilience to omission bias. A natural diagnostic – computable from the estimated whitened matrix $\hat{W}$ prior to performing any structural inference – consists of examining the minimum spectral gap $\min_j \hat{\delta}_j = \min_j (\hat{\mu}_j - \hat{\mu}_{j+1})$. A small minimum gap signals that at least one structural direction is potentially fragile to omission bias, even in the absence of direct evidence of misspecification (although notice that the DK bound uses the true spectral gaps, not the estimated ones, so this is only an approximative diagnostic).

The Error Decomposition

To conclude this simulative section, the final experiment tests Proposition 2.2 by measuring the contributions of each term in the error decomposition (2.30) separately for our DGP. Recall that the identification error admits the exact algebraic decomposition of Proposition 2.2. For each Monte Carlo replication, we compute the Frobenius norm of each term individually: $\|(\hat{C} - \tilde{C})\tilde{Q}\|_F$, $\|\tilde{C}(\hat{Q} - \tilde{Q})\|_F$, and $\|(\hat{C} - \tilde{C})(\hat{Q} - \tilde{Q})\|_F$. We stress that these three norms do not in general sum to $\|\hat{B} - \tilde{B}\|_F$ (as their sum is an upper bound by the triangle inequality (Axler, 2024)), but the three quantities being averaged are components of the exact identity (2.2). Averaging them separately across replications, thus gives the expected

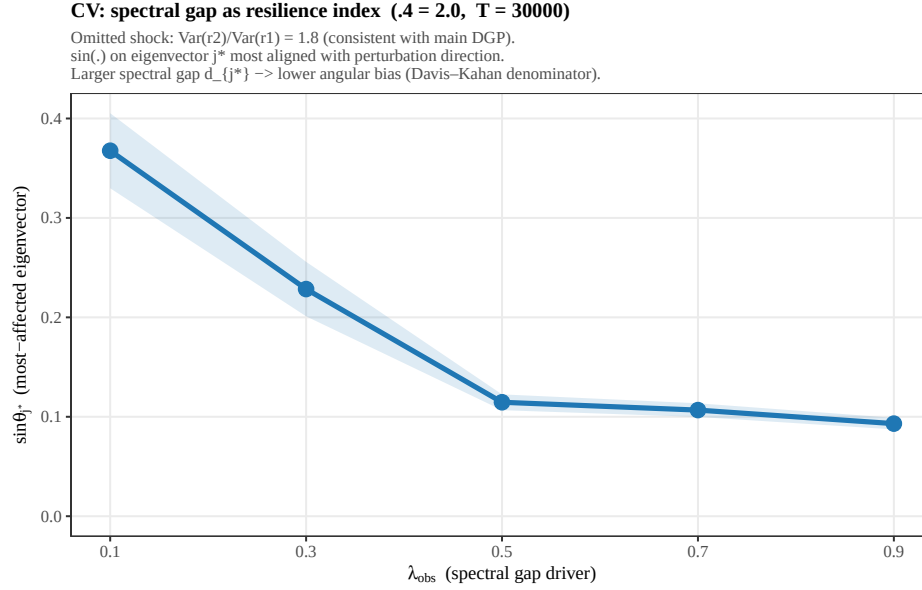


Fig. 7: Asymptotic angular bias $\sin \Theta(\hat{q}_{j^*}, \tilde{q}_{j^*})$ as a function of the spectral gap driver λ_{obs} .

magnitude of each contribution. Figure 8 presents the mean Frobenius norm of each term as a stacked bar chart, across three contamination levels $\lambda_4 \in \{2, 4, 8\}$ and four sample sizes $T \in \{300, 1,000, 3,000, 10,000\}$. The decomposition reveals a sharp contrast between the two channels.

Channel 1 (Cholesky distortion). The mean norm $\mathbb{E}\|(\hat{C} - \tilde{C})\tilde{Q}\|_F$ is approximately constant across the three panels at any given sample size, and declines steadily as T increases. The λ_4 -invariance is consistent with the theoretical argument: $\hat{C}$ is determined by the contaminated regime-1 covariance $\hat{\Sigma}^{(1)} = \tilde{B}\tilde{B}^T + P_1$, and $P_1 = cc^{T15}$ depends on the regime-1 variance of the omitted shock (which is 1 under the normalisation $\Lambda^{(1)} = I$) but not on λ_4 . As a consequence, the asymptotic Cholesky error is the same regardless of how volatile the omitted variable is in regime 2. The decline with T reflects the standard $\sqrt{T}$ -consistency of the sample covariance estimate of $\hat{\Sigma}^{(1)}$; in the limit, $\hat{C} \rightarrow C^* \neq \tilde{C}$, so the Channel 1 contribution stabilises at a positive constant for large T .

Channel 2 (eigenvector rotation). The mean norm $\mathbb{E}\|\tilde{C}(\hat{Q} - \tilde{Q})\|_F$ increases markedly from left to right as λ_4 grows. At $\lambda_4 = 2$, Channel 2 is larger than Channel 1 at $T = 300$ and becomes comparable at $T = 3,000$. At $\lambda_4 = 8$, Channel 2 dominates everywhere, accounting

¹⁵ To maintain the design simple, we retain only the contemporaneous part pf the perturbation, which accounts for the biggest share of total magnitude.

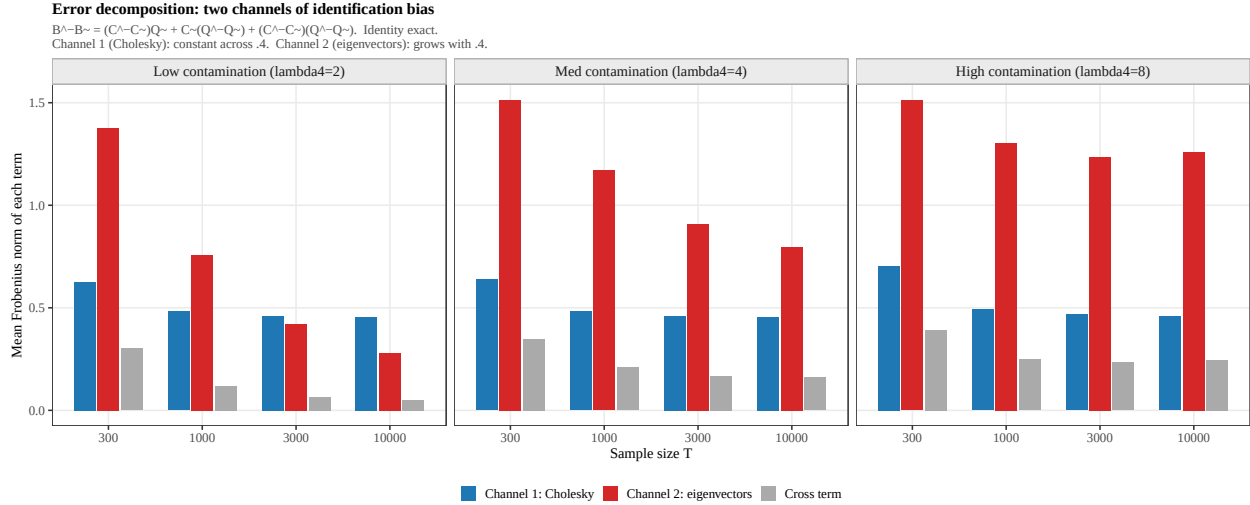


Fig. 8: Mean Frobenius norm of each term in the error decomposition.

for the largest share of the total identification error at every sample size. This ordering is theoretically expected: the eigenvector rotation is bounded by $\|E_W\|_{\text{op}}/\delta_j$, and $\|E_W\|_{\text{op}}$ grows with λ_4 whereas Channel 1 is λ_4 -invariant. The cross term $\|(\hat{C} - \tilde{C})(\hat{Q} - \tilde{Q})\|_F$ is small relative to both channels in all panels, highlighting that, at least for this DGP, a two-channel decomposition provides an adequate account of the total error.

Taken together, the decomposition results support the following practical reading of the theory. At moderate contamination levels, both channels contribute comparably to the total identification error, and the researcher should be concerned about both the distortion of the whitening step (Channel 1) and the rotation of the structural directions in the whitened space (Channel 2). At high contamination, Channel 2 dominates: the eigenvector rotation becomes the primary source of failure. This suggests that, in applications where the omitted variable is suspected of being highly heteroskedastic, the spectral gap of the estimated $\hat{W}$ matrix is likely to be the most informative single diagnostic for the robustness of structural identification.

4 Empirical Application

To conclude our exposition, we now present a brief empirical analysis, partly inspired - in context and spirit - by De Arcangelis and Di Giorgio (2000). Indeed, they estimate different VAR models by adding some new variables in the process (the so-called *nested models*) as a robustness check. This is directly relevant to our discussion, and we borrow the idea by analysing Italian macro data from 1999M1 to 2019M12. However, while their main purpose is the identification of monetary policy shocks, alongside the behaviour of the other variables when separately hit by each one of these shocks, our interest resides in discovering to which extent the findings discussed in the previous, theoretical sections, ultimately apply to a standard empirical situation.

To fruitfully apply our CV-SVAR methodology, we estimate three nested models of increasing dimension: a baseline specification **M1** including a consumer price index, an industrial production index (excluding the construction sector), and the series of the ECB's MRO (Main Refinancing Operation) rate; a four-variable extension **M2** which adds the BTP/Bund yield spread, and a five-variable benchmark **M3** that further includes banking credit to the private sector¹⁶. All models are estimated as a VAR(6) with a constant intercept and 3 lags chosen via AIC information criterion (see Section 1 for the details), consistent with standard practice.

M1 (3-var): $\log(\text{HICP}) \times 100$, $\log(\text{IP}) \times 100$, MRO.

M2 (4-var): M1 + BTP/Bund 10-year yield spread (percentage points).

M3 (5-var): M2 + $\log(\text{banking credit}) \times 100$ (interpolated from quarterly).

Of course, CV identification requires a well-specified structural break. Mario Draghi's *whatever it takes* speech of July 2012 and the announcement of Outright Monetary Transactions provide a natural and recognized regime shift (Di Giorgio, 2024). Regime 1 covers January 1999 to June 2012 (high volatility, sovereign debt crisis); Regime 2 covers July 2012 to December 2019 (low volatility, post-OMT normalisation). The break is also validated by a Chow test for the benchmark model **M3**, following Lanne and Lütkepohl (2008). Although a simple Portmanteau test rejects the null hypothesis of zero residuals autocorrelation in

¹⁶ All time series are downloaded from FRED: **ITACPIALLMINMEI**, **ITAPROINDMISMEI**, **ECBMRRFR**, **IRLTLT01ITM156N**, **IRLTLT01DEM156N**, **CRDQITBPABIS**. They are then converted to monthly frequency via the *approx()* function of the **stats** package in R, aligned and trimmed to match the desired temporal window.

Tab. 1: Operator-norm decomposition of the identification error.

Comparison	$\ \Delta B\ _{\text{op}}$	$\ C_1\ _{\text{op}}$	$\ C_2\ _{\text{op}}$	$\ C_3\ _{\text{op}}$	Dominant Channel
M1 vs M2 (omitted: Spread)	0.19989	0.03733	0.20380	0.00243	C_2 (eigenvector)
M1 vs M3 (omitted: Spread, Credit)	0.19054	0.05397	0.17993	0.00678	C_2 (eigenvector)
M2 vs M3 (omitted: Credit)	0.05350	0.03512	0.04741	0.00121	C_2 (eigenvector)

Note: The three norms do not in general sum to $\|\Delta B\|_{\text{op}}$ due to the triangle inequality, but $C_1 + C_2 + C_3 = \Delta B$ holds exactly by definition.

all three specifications, we argue that this does not invalidate our application because our aim is not to build a definitive SVAR model to conduct causal inference, but instead the objective is to apply the DK framework to the nested models procedure described above. In accordance to the procedure of Section 2.1, for each model we compute the sample covariance matrices $\hat{\Sigma}^{(1)}$ and $\hat{\Sigma}^{(2)}$ from the VAR residuals in each regime, obtain the Cholesky factor $\hat{C}$ of $\hat{\Sigma}^{(1)}$, construct the whitened matrix $\hat{W} = \hat{C}^{-1}\hat{\Sigma}^{(2)}\hat{C}^{-T}$, and recover the structural impact matrix as $\hat{B} = \hat{C}\hat{Q}$, where $\hat{Q}$ collects the eigenvectors of $\hat{W}$. The three recovered eigenvalues $\hat{\mu}_j$ of $\hat{W}$ measure the volatility ratio of shock j between regimes. For **M3** these are $\hat{\mu} = (1.98, 0.76, 0.35)$, indicating that Shock 1 approximately doubles its variance in Regime 2 while Shocks 2 and 3 attenuate. The central exercise of this section is to treat **M3** as the full-information benchmark and examine how the CV estimator behaves when the researcher omits one or two variables. Thus, the 'true' values of the quantities aforementioned, needed to compute the diagnostics presented in this work, are extracted from the shared 3×3 subsystem. The three pairwise comparisons are performed, and for each comparison, the Davis–Kahan diagnostics of Section 2.4 are computed.

Table 1 reports the operator-norm decomposition of the identification error $\hat{B} - \tilde{B}$ into its two structural channels (Proposition 2.2). Channel 2 (eigenvector rotation, C_2) dominates in all three comparisons. This finding is consistent with the Monte Carlo evidence: at empirically plausible contamination levels, the eigenvector rotation driven by the perturbation of $\hat{W}$ is the primary source of identification failure. Table 2 presents the full Davis–Kahan diagnostics. A key distinction is the one between the whitened-space angle θ_j^W and the original-space angle θ_j^B . The DK bound of Lemma 2.3 governs $\sin \theta_j^W$, and is satisfied in all nine shock-comparison cells. In the whitened space, the prediction of the theory is confirmed monotonically: Shock 1, with the largest spectral gap $\delta_1 \approx 1.27\text{--}1.32$, exhibits the smallest whitened-space rotation ($\theta_1^W \leq 2.1^\circ$ across all comparisons), while Shocks 2 and 3

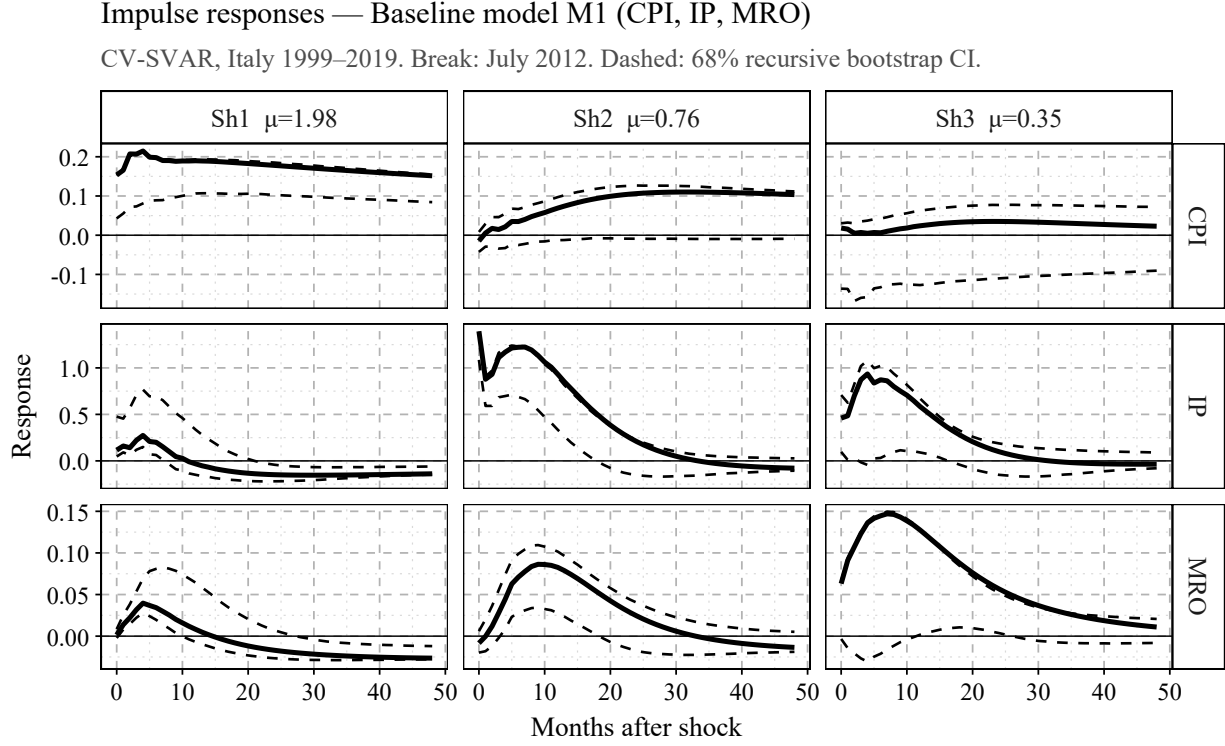


Fig. 9: Impulse Response Functions for the baseline specification.

($\delta_{2,3} \approx 0.40\text{--}0.44$) rotate by $7\text{--}8^\circ$. The bound uses only 2–16% of its capacity for Shock 1 versus up to 30% for Shocks 2–3, consistent with the geometric prediction that a larger spectral gap provides stronger structural resilience. For completeness, Table 3 reports the structural impact matrices B of the 3×3 subsystem estimated across the three different specifications. The $\hat{B}$ of **M3** should be treated as $\tilde{B}$, even though it probably suffers from more variable omission.

Impulse Response Analysis

The figures that follow report the impulse responses of CPI, IP, and MRO to each identified shock, across models **M1** – **M3**. Inference on the IRFs is conducted via a residual-based, recursive-design bootstrap, stratified by volatility regime (Lütkepohl and Kilian, 2017, Sec. 12.2.1). Let $\hat{U}^{(k)} = \{\hat{u}_t^{(k)}\}_{t=p+1}^T$ be the $(T-p) \times K_k$ matrix of OLS residuals.¹⁷ Denote by τ the break index in the residual space, and let $\mathcal{I}_1 = \{1, \dots, \tau-1\}$ and $\mathcal{I}_2 = \{\tau, \dots, T-p\}$ be the two regime index sets, with n_r the number of observations per regime after deleting

¹⁷ Here, $p = 6$ denotes the lag order, while K_k the dimension of model **Mk**, with $K_1 = 3$, $K_2 = 4$, $K_3 = 5$.

the first p observations. For each bootstrap replication $b = 1, \dots, B$, the workflow is as follows:

1. Draw $s_1^{(1)}, \dots, s_{n_1}^{(1)}$ with replacement from $\mathcal{I}_1$ and, independently, $s_1^{(2)}, \dots, s_{n_2}^{(2)}$ with replacement from $\mathcal{I}_2$. This generates what is technically referred to as the *empirical distribution*. Join the two sampled vectors to form the index vector $\mathbf{s} = (s_1^{(1)}, \dots, s_{n_1}^{(1)}, s_1^{(2)}, \dots, s_{n_2}^{(2)})$.
2. Simulate the bootstrap sample recursively, holding the first p observations fixed at their observed values (Lütkepohl and Kilian, 2017, p. 344):

$$y_t^{(k),*} = \hat{\nu}^{(k)} + \sum_{\ell=1}^p \hat{A}_\ell^{(k)} y_{t-\ell}^{(k),*} + \hat{u}_{s_{t-p}}^{(k)}, \quad t = p+1, \dots, T,$$

where $\hat{\nu}^{(k)}$ and $\hat{A}_\ell^{(k)}$ are the estimated VAR intercept and lag matrices.

3. Re-estimate the VAR on the full time-series vector $Y_b^{(k),*}$ and apply `id.cv` at the same break point τ to obtain $\hat{B}_b^{(k),*}$; align its columns to $\hat{B}^{(1)}$.
4. Compute the bootstrap IRF confidence bands based on the different quantiles of estimated impact vectors.

Stratifying the resampling by regime preserves the cross-regime variance contrast that `id.cv` exploits for identification. In the baseline figure the band for **M1** is shown as dashed lines; in the comparison figures only the **M3** band is displayed (grey ribbon), so that the point estimates of **M1** and **M2** can be assessed against the sampling uncertainty of the benchmark model. In the baseline figure (Figure 9), the 68% band for **M1** is displayed as dashed lines around the point estimate. In the model-comparison figures which follow, only the 68% band of the benchmark model is shown (grey shaded ribbon), against which the point estimates of **M1** and **M2** are plotted. We thus report uncertainty for the best-specified model and use this band as a reference zone for assessing whether the nested models' IRFs are statistically distinguishable from **M3**. A nested model's point estimate falling outside the **M3** ribbon is interpretable as evidence that the omitted variable induces a bias in the structural impact matrix that exceeds sampling uncertainty. This, for instance, is the case of the responses of CPI to the first and third shock.

Shock 1 — CPI shock ($\hat{\mu}_1 = 1.98$). A positive realisation of Shock 1 produces a persistent rise in CPI, that shows no sign of inversion within the 48-month horizon except for the more complete specification. Industrial production responds positively on impact - perhaps as a

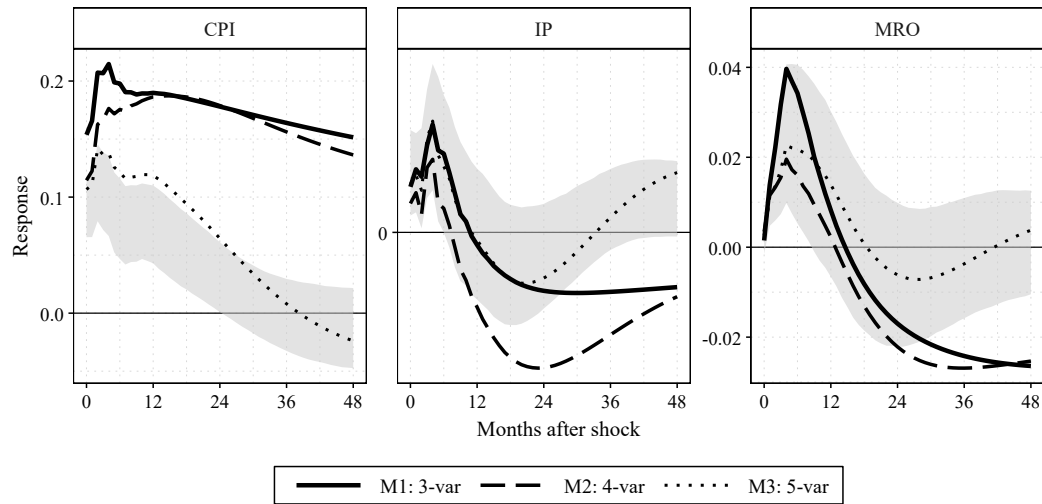
consequence of the higher prices - but turns negative after approximately twelve months. The MRO rises, consistently with a systematic monetary policy reaction to above-target inflation. The combination of durable price pressure, medium-run output contraction, and endogenous tightening is the hallmark of a cost-push or supply-side disturbance.

Shock 2 — Production (demand?) shock ($\hat{\mu}_2 = 0.76$). Shock 2 generates the cleanest pattern of the three. IP responds sharply on impact, and returns to baseline within 24 months. CPI rises gradually and persistently, reflecting the transmission of demand pressure on prices through nominal rigidities. MRO traces a hump-shaped response, peaking around 12 months before reverting - conformably with a delayed but transitory tightening of monetary policy in response to the demand impulse.

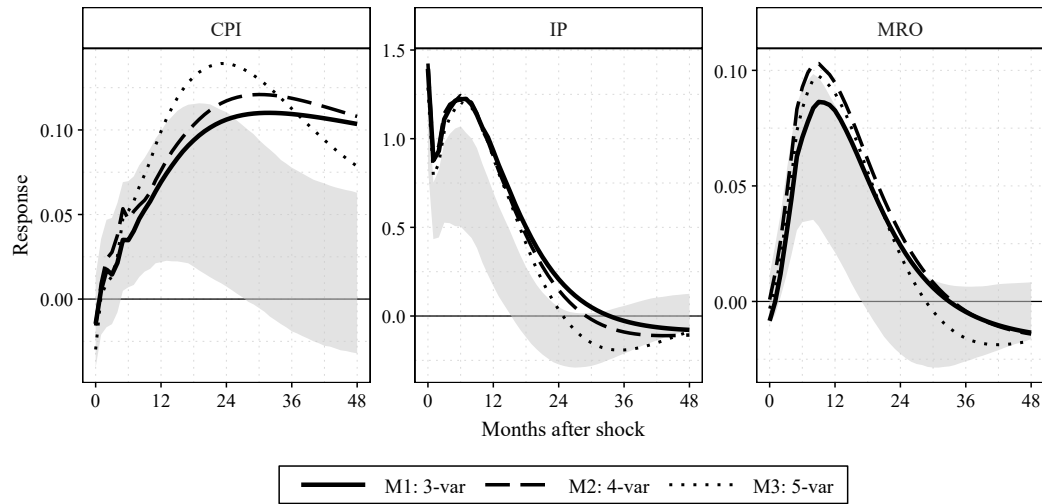
Shock 3 — MRO (monetary policy) shock ($\hat{\mu}_3 = 0.35$). The responses to Shock 3 present a well-known identification challenge. MRO rises on impact while both CPI and IP rise, a pattern inconsistent with the conventional view of a contractionary monetary shock. This constitutes an instance of the *price puzzle* and *output puzzle* documented extensively in the SVAR literature (Sims, 1992). The anomaly is likely due to information omission: in the absence of financial variables, the VAR cannot distinguish an exogenous rate change from an endogenous response to inflationary signals observed by the central bank but absent from the model. This explains also why the inclusion of more variables in **M3** seems to provide more sensible results. It is essentially, once again, a problem of variable omission and insufficient information.

Impulse responses to Sh1 ($\mu=1.98$)

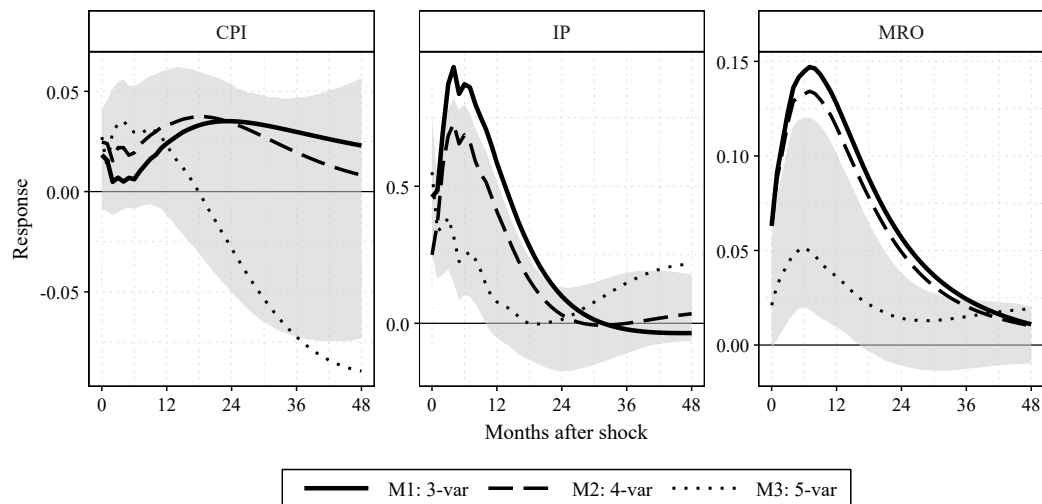
CV-SVAR, Italy 1999–2019. Shaded: 68% recursive bootstrap CI of benchmark M3.

Impulse responses to Sh2 ($\mu=0.76$)

CV-SVAR, Italy 1999–2019. Shaded: 68% recursive bootstrap CI of benchmark M3.

Impulse responses to Sh3 ($\mu=0.35$)

CV-SVAR, Italy 1999–2019. Shaded: 68% recursive bootstrap CI of benchmark M3.



Tab. 2: Davis–Kahan diagnostics for the three nested-model comparisons.

Comparison	Shock	$\tilde{\mu}_j$	δ_j	$\ E_W\ _{\text{op}}$	Bound DK(°)	$\theta_j^W(°)$	$\theta_j^B(°)$	θ_j^B/θ_j^W	DK
M1 vs M2									
(omitted: Spread)	$j = 1$	2.0186	1.3228	0.1172	10.21	0.77	3.98	5.17x	✓
	$j = 2$	0.6957	0.4035		35.53	8.04	0.39	0.05x	✓
	$j = 3$	0.2923	0.4035		35.53	8.02	5.97	0.74x	✓
M1 vs M3									
(omitted: Spread, Credit)	$j = 1$	2.0136	1.2680	0.0968	8.78	1.57	11.53	7.35x	✓
	$j = 2$	0.7455	0.4389		26.17	7.44	0.29	0.04x	✓
	$j = 3$	0.3067	0.4389		26.17	7.40	5.35	0.72x	✓
M2 vs M3									
(omitted: Credit)	$j = 1$	2.0136	1.2680	0.0776	7.03	2.07	7.56	3.66x	✓
	$j = 2$	0.7455	0.4389		20.72	2.03	0.21	0.10x	✓
	$j = 3$	0.3067	0.4389		20.72	1.04	0.69	0.66x	✓

Note: ✓ indicates that the DK bound is respected in the whitened space ($\theta_j^W \leq \text{Bound}^\circ$) — satisfied in all nine cases. The amplification factor for Shock 1 (4–7×) reflects the differential stretching of that structural direction by the Cholesky factor.

Tab. 3: Estimated structural impact matrices $\hat{B}$ for the three nested models.

	M1: 3-var			M2: 4-var			M3: 5-var (benchmark)		
	$j = 1$	$j = 2$	$j = 3$	$j = 1$	$j = 2$	$j = 3$	$j = 1$	$j = 2$	$j = 3$
CPI	0.1537	-0.0147	0.0183	0.1498	-0.0098	0.0203	0.1439	-0.0145	0.0214
IP	0.1157	1.3940	0.4600	0.1300	1.4307	0.2640	0.1627	1.3899	0.2754
MRO	0.0015	-0.0084	0.0631	0.0024	-0.0003	0.0631	0.0020	-0.0013	0.0623

5 Conclusions

We conclude our work with a short appraisal of its main contributions and limitations. On the positive side, we managed to establish theoretically the generic inconsistency of the CV identification method under the omission of at least one relevant variable. The bias introduced was *exactly* decomposed in two qualitatively different channels – alongside their dynamic interaction –, one related to the whitening distortion prior to simultaneous diagonalization of the covariance matrices, and one specifically tied to the incorrect retrieval of eigenvectors in the whitened space. Moreover, the introduction to the CV-SVAR literature of the Davis–Kahan bound is genuinely new and offers a fresh perspective on the dynamics and the determinants of correct identification. The spectral gap, in particular, seems to constitute an index of identifying power as well as of misspecification robustness. The Monte Carlo results validate each one of the relevant theoretical propositions: asymptotic plateau at B^* , dominance of Channel 2 under consistent volatility of the omitted shock, monotonicity with respect to the magnitude of the perturbation and the length of the spectral gap. A short empirical illustration applies and validates the bound in a concrete macroeconomic setting. With all these successes, comes also a number of limitations. Our treatment developed only the bi-regime case and, although the extension to a multi-regime setting should be algebraically straightforward, it remains to be formalised. More relevantly, the Davis–Kahan bound, although useful to understand the dynamics of the perturbation, has often proved to be far too conservative to be useful as a parsimony filter, that is, to discriminate between superfluous and essential time series. Lemma A.2 in the Appendix allows for rather small perturbation magnitudes, advocating for the inclusion even of variables theoretically unlinked to the system considered. This limits its employability as an empirical tool, even though it does not diminish its validity as a theoretical and heuristic device.

With these considerations in mind, a few possible directions of future research emerge: first of all, finding a way to “tighten” the bound, perhaps exploiting the DGP-specific features of the perturbation (e.g., a directional bound which accounts not for the worst-case scenario but instead takes into account the specific direction of the perturbation in the whitened system). Another interesting extension would be to deepen the analysis through the examination of multiple breaks and/or stochastic volatility, where the number of regimes itself is uncertain. Finally, it might be possible to develop a formal testing procedure by comparing the differences among the nested whitened matrices with the critical perturbation expression found through the tighter version of the bound. Notwithstanding these limitations, the

present work aims to offer a formal contribution to the understanding of the spectral frailty of CV identification, in the hope that this "perturbative" perspective opens new pathways of empirical and theoretical research.

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A Supporting Lemmas and Assumptions

Lemma A.1 (Vanishing of cross-terms). *Let $\tilde{u}_t = \tilde{B}\tilde{w}_t$ be the residual of the misspecified VAR and let η_t denote the contamination term arising from the omitted variable as in (2.16). Then*

$$\mathbb{E}[\tilde{u}_t \eta_t^T] = 0.$$

Proof. Decompose η_t into its contemporaneous and lagged components.

(i) *Contemporaneous part.* The term η_t contains the omitted structural shock w_{4t} . By the SVAR assumption of mutually uncorrelated structural disturbances, $\mathbb{E}[\tilde{w}_t w_{4t}^T] = 0$, and since $\tilde{u}_t$ is a linear function of $\tilde{w}_t$, it follows that $\mathbb{E}[\tilde{u}_t (c w_{4t})^T] = 0$.

(ii) *Lagged part.* The remaining components of η_t are functions of $y_{4,t-s}$ and $\tilde{y}_{t-s}$ for $s \geq 1$. Because $\tilde{u}_t$ is the one-step-ahead innovation of a stationary VAR, it is uncorrelated with the entire information set available at $t - 1$ by the white noise assumptions:

$$\mathbb{E}[\tilde{u}_t y_{t-s}^T] = 0 \quad \text{for all } s \geq 1.$$

Combining (i) and (ii) gives $\mathbb{E}[\tilde{u}_t \eta_t^T] = 0$. □

Lemma A.2 (Critical perturbation and structural resilience). *Under the assumptions of Lemma 2.3, fix a tolerance $\sin \theta^* \in (0, 1)$. The largest perturbation magnitude consistent,*

over all possible directions of E_W , with a rotation of whitened axis j below θ^* is

$$\|E_W\|_{\text{op}}^* = \frac{\delta_j \sin \theta^*}{2}.$$

The spectral gap δ_j enters linearly: a system with twice the gap can tolerate twice the contamination for the same worst-case accuracy.

Proof. Inverting the bound (2.35) of Lemma 2.3 at the tolerance $\sin \theta^*$ directly yields $\|E_W\|_{\text{op}} \leq \delta_j \sin \theta^* / 2$. $\square$

As extensively discussed in the previous sections, this critical perturbation accounts for the worst possible contamination scenario: indeed, it is possible for the system to experience significantly more serious perturbations without exceeding the chosen tolerance.

Lemma A.3 (Channel 2 vanishing). *Under the assumptions of Proposition 2.1, $\hat{Q} = \tilde{Q}$ if and only if $E_W, \tilde{W}$ commute.*

Proof. ($\Leftarrow$) First, suppose $E_W, \tilde{W}$ commute. For each j , since $\tilde{W}\tilde{q}_j = \mu_j\tilde{q}_j$,

$$\tilde{W}(E_W\tilde{q}_j) = E_W(\tilde{W}\tilde{q}_j) = \mu_j(E_W\tilde{q}_j),$$

so $E_W\tilde{q}_j$ is an eigenvector of $\tilde{W}$ with eigenvalue μ_j . Since the μ_j are distinct, each eigenspace is one-dimensional, hence $E_W\tilde{q}_j = \alpha_j\tilde{q}_j$ for some scalar α_j . Therefore $\hat{W}\tilde{q}_j = (\mu_j + \alpha_j)\tilde{q}_j$, so $\tilde{Q}$ diagonalises $\hat{W}$ and $\hat{Q} = \tilde{Q}$.

($\Rightarrow$) If $\hat{Q} = \tilde{Q}$, both $\tilde{W}$ and $\hat{W} = \tilde{W} + E_W$ are diagonal in the basis $\{\tilde{q}_j\}$, hence so is E_W . A matrix diagonal in an orthonormal basis commutes with any other matrix diagonal in the same basis, proving the statement in the other direction. $\square$

Assumption A.1 (Non-degenerate loading structure). *The matrix*

$$C = [c \mid a_1 \mid \cdots \mid a_p \mid B_1 \mid \cdots \mid B_p] \in \mathbb{R}^{n \times (p+1+np)}$$

has full row rank, $\text{rank}(C) = n$.

Here $c \in \mathbb{R}^n$ is the contemporaneous impact of the omitted variable, $a_s \in \mathbb{R}^n$ are its lag-propagation vectors up to p , and $B_s = \hat{A}_s - \tilde{A}_s \in \mathbb{R}^{n \times n}$ are the coefficient bias matrices arising from the misspecified projection on the included variables. The contamination term can then be written as

$$\eta_t = C \zeta_t, \quad \zeta_t = (w_{4t}, y_{4,t-1}, \dots, y_{4,t-p}, \tilde{y}_{t-1}^T, \dots, \tilde{y}_{t-p}^T)^T,$$

so that $P_k = C \Omega_k C^T$ with $\Omega_k = \mathbb{E}[\zeta_t \zeta_t^T \mid k] \succ 0$. Under Assumption A.1, P_k is SPD since C has full row rank and $\Omega_k \succ 0$ by construction. The condition requires that the combined set of loading vectors — contemporaneous impact, lag-propagation of the omitted variable, and coefficient bias directions — is not collinear, i.e. that the omitted variable contaminates the system along sufficiently distinct directions at different horizons. This is generically satisfied: for a stationary omitted process with non-trivial autoregressive dynamics, the impulse response vectors at successive lags will almost surely span $\mathbb{R}^n$.

Remark. Even when Assumption A.1 holds, P_k may be nearly singular if $\|c\| \gg \|a_s\|$ for all $s \geq 1$ and the bias matrices B_s are small, in which case $P_k \approx \sigma_{w_4}^2 c c^T$ is approximately rank-one and ill-conditioned. Any quantity involving $P_1^{-1} P_2$ then becomes numerically unreliable. In such an instance, the perturbation impacts the system effectively in a single direction.